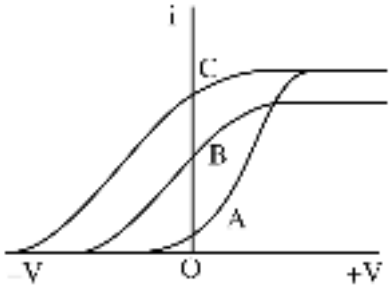
PHYSICS

1) **Assertion (A) :** Minimum energy required for an electron in metal so that it can escape from surface.
Reason (R) : As electron attempts to escape metal surface aquires positive charge.

(1) Both (A) and (R) are correct but (R) is correct explanation of (A)
(2) Both (A) and (R) are correct and (R) is not correct explanation of (A)
(3) Both (A) & (R) incorrect
(4) (A) incorrect while (R) correct

2) The following graph is between anode potential and photoelectric current in photoelectric experiment for three different light radiations than :-



(1) Wavelength of A is less than that of B
(2) Wavelength of C is maximum while that of A is minimum
(3) Wavelength of B is more than C but less than A
(4) Intensity of A is less than C

3)

A source of power 15 mW is emitting light of wavelength 440 nm. If source capable to convert 10% electrical energy into light then number of quanta emitted by the source per minute.

(1) 1.98×10^{17}
(2) 3.2×10^{15}
(3) 5.4×10^{10}
(4) 1.6×10^{12}

4) Light of wavelength 500 nm incident on photocathode in photoelectric effect experiment. If work function is 1.2 eV and anode is kept at 4 volt. Find kinetic energy of fastest and slowest electrons reaching anode :-

(1) 6.28 eV and 3 eV
(2) 5.28 eV and 4 eV
(3) 3.52 eV and 4 eV

(4) 1.42 eV and 0.5 eV

5) A point source of light is 2m apart from a photocell, saturation current and stopping potential found to be 1.4 μA and 0.5 volt if the separation between the source and photocell is increased to 6 m then saturation current and stopping potential will be :-

(1) 0.15 μA, 0.5 V

(2) 0.30 μA, 1.5 V

(3) 4.5 μA, 0.2 V

(4) 0.5 μA, 3V

6) In hydrogen atom transition of electron takes place from energy level n = 4 to energy level n = 2 therefore a photon is emitted, find recoil speed of the atom, m = mass of H atom, R = Rydberg constant, h = plank's constant :-

(1) $\frac{3Rh}{16m}$

(2) $\frac{5Rh}{3m}$

(3) $\frac{2Rh}{3m}$

(4) $\frac{15}{16}\frac{Rh}{m}$

7) Photon of wavelength 500 nm incident on a metal of work function 2.44 eV. The de-Broglie wavelength of emitted photo electron will be :-

(1) = 6.135×10^{-11} m

(2) ≥ 6.135×10^{-9} m

(3) < 6.135×10^{-12} m

(4) ≤ 6.135×10^{-12} m

8) Energy of a photon whose de Broglie wavelength is equal to the wavelength of an electron accelerated through a potential difference of 125 V is near to

(1) 11.5 eV

(2) 11.5 keV

(3) 125 eV

(4) 1250 eV

9) Potential energy of a particle changes along x axis as $\begin{bmatrix} 0 \leqslant x \leqslant 1 & E_0 \\ x > 1 & \frac{E_0}{2} \end{bmatrix}$ than ratio of de-Broglie wavelengths of this particle in the region of $0 \leq x \leq 1$ and $x > 1$. If total mechanical energy of the particle is $2E_0$:-

(1) $\sqrt{3} : \sqrt{2}$

(2) $\sqrt{3} : 1$

(3) $1 : \sqrt{3}$

(4) 2 : 1

10) An element A decays into an element C by two step process

$A \rightarrow B + {}_{2}^{4}He$

$B \rightarrow C + 2\ e^{-}$

Then

(1) A and C are isobars

(2) A and C are isotopes

(3) A and B are isotopes

(4) A and B are isobars

11) What is the energy released in process

$$3{}_{2}^{4}He \rightarrow {}_{6}^{12}C \left[\begin{array}{l} \text{mass of } {}_{2}^{4}He = 4.002604\,u \\ \text{mass of } C^{12} = 12.00000\,u \end{array} \right]$$

(1) 7.2 MeV

(2) 9.38 MeV

(3) 6.09 MeV

(4) 10.9 MeV

12) A electron beam of 12.5 eV is used to bombard gaseous hydrogen at room temperature, what series of wavelengths will be emitted :-

(1) Lymen, Balmer

(2) Balmer only

(3) Lymen only

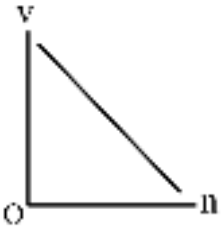
(4) Neither Lymen nor Balmer

13) Which of the following curves may represent the speed of the electron in a hydrogen atom as a function of the principal quantum number n :-

(1)



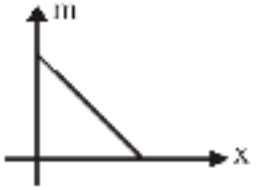
(2)

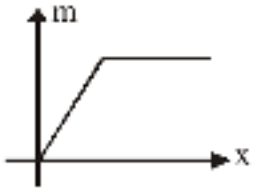


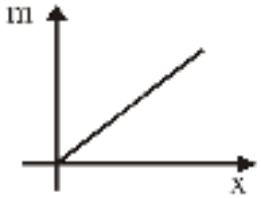
(3) 

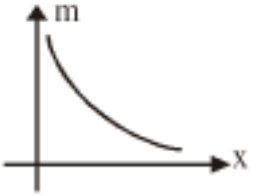
(4) 

14) Which of the following graphs is the magnification of a real image against the distance of object from the focus of a concave mirror?

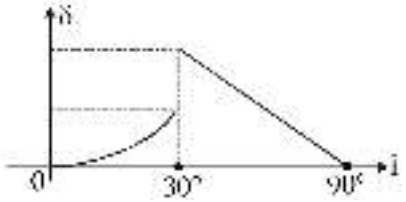
(1) 

(2) 

(3) 

(4) 

15) Figure shows graph of angle of deviation v/s angle of incidence for a light ray. Incident ray goes from medium 1 (μ_1) to medium 2 (μ_2). Calculate the maximum deviation.



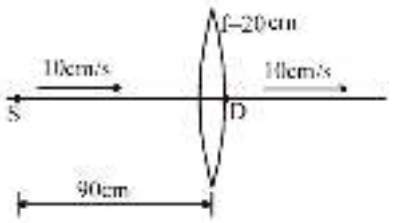
(1) 60°

(2) 90°

(3) 120°

(4) 150°

16) The figure shows initial positions of a source of light S and a light detector D. Both S & D move with velocity 10 cm/s, as shown. The converging lens of focal length 20 cm is fixed. Time instant(s)



when the detector detects maximum intensity of light is /are :-

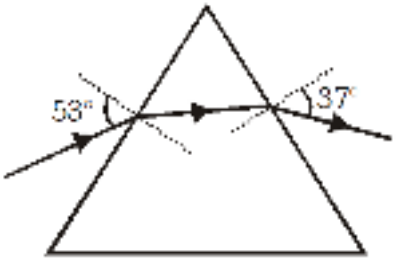
(1) 3 sec

(2) 6 sec

(3) equal intensity at all time

(4) Both 1 & 2

17) A ray incident at an angle 53° on a prism emerges at an angle 37° as shown. If the angle of incidence is made 50°, which of the following is a possible value of the angle of emergence ?



(Minimum deviation occurs when angle of incidence is less than 45°)

(1) 35°

(2) 42°

(3) 40°

(4) 38°

18) A man is looking at a small object placed at his near point. Without altering the position of his eye or the object, he puts a simple microscope of magnifying power 5X before his eyes. The angular magnification achieved is

(1) 5

(2) 2.5

(3) 1

(4) 0.2

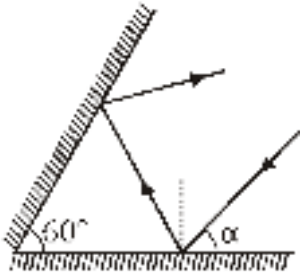
19) The focal length of the objective of a compound microscope is f_0 and its distance from the eyepiece is L. The object is placed at a distance u from the objective. For proper working of the instrument.

(1) $L < u$

(2) $L < f_0$

(3) $f_0 < l < 2f_0$

(4) $L > 2f_0$

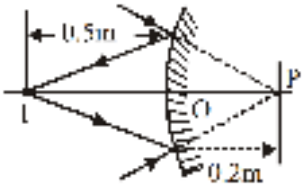
20) What is the value of α for which there are maximum 2 reflections only ? 

(1) 35°

(2) 45°

(3) 55°

(4) 65°

21) A beam of convergent light converges to a point 0.5 m in front of the mirror after reflection at a convex mirror but in the absence of the mirror the beam converges to a point 0.2 m behind the mirror. The radius of curvature of the mirror is :- 

(1) 20 cm

(2) 50 cm

(3) 66.67 cm

(4) 28.57 cm

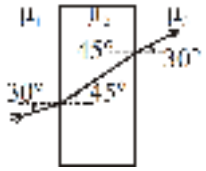
22) At what angle with the normal (in degree) should a light beam fall on the surface from air to material of refractive index n = 1.732, if the angle of refraction was half the angle of incidence? Assume that the beam is not incident normally.

(1) 60°

(2) 45°

(3) 30°

(4) 15°

23) Figure shows a ray of light passing through a glass slab. From figure, choose the correct option :- 

(1) $\mu_1 = \mu_2$

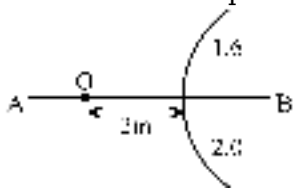
(2) $\mu_1 < \mu_2$

(3) $\mu_2 < \mu_3$

(4) $\mu_1 < \mu_3$

24) In the figure shown a point object O is placed in air. A spherical boundary of radius of curvature

1.0 m separates various media. AB is principal axis. The refractive index above AB is 1.6 and below AB is 2.0. The separation between the images formed due to refraction at spherical surface is :



(1) 12 m

(2) 20 m

(3) 14 m

(4) 10 m

25) You are given two lenses, a converging lens with focal length + 10 cm and a diverging lens with focal length – 20 cm. Which of the following would produce a virtual image that is larger than the real object ?

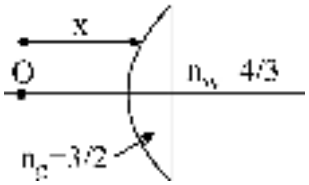
(1) Placing the object 5 cm away from the converging lens.

(2) Placing the object 15 cm away from the converging lens.

(3) Placing the object 5 cm away from the diverging lens.

(4) Placing the object 15 cm away from the diverging lens.

26) An object 'O' is kept in air in front of a thin plano convex lens of radius of curvature 10 cm. Its refractive index is 3/2 and the medium towards right of plane surface is water of refractive index 4/3. What should be the distance 'x' of the object so that the rays become parallel finally?



(1) 5 cm

(2) 10 cm

(3) 20 cm

(4) None of these

27) A convex lens is placed between an object and a screen which are at a fixed distance D apart for one position of the lens. The magnification of the image obtained on the screen is $m_1 = 2$. When the lens is moved by a distance d = 30 cm the magnification of the image obtained on the same screen is $m_2 = 0.5$. Find the focal length of the lens.

(1) 12 cm

(2) 20 cm

(3) 15 cm

(4) 45 cm

28) An astronomical telescope has magnifying power 6 for distant objects. The separation between objective and the eye piece is 42 cm and the final image is formed at infinity. The focal length f_0 of objective and focal length f_e of eye-piece are (in cm) :-

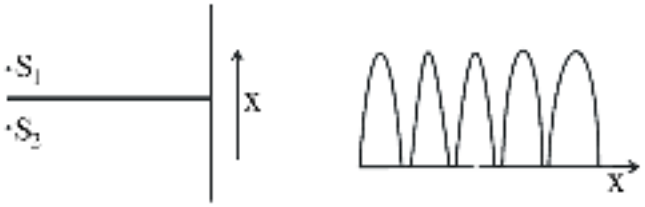
(1) $f_0 = 36, f_e = 6$

(2) $f_0 = 50, f_e = 8$

(3) $f_0 = 20, f_e = 10$

(4) $f_0 = 40, f_e = 8$

29) In a standard YDSE setup, two source S_1 and S_2 of intensity I_1 and I_2 are placed in front of a screen (figure). The pattern of intensity distribution seen in the central portion is given by graph.In this case which of the following statements are not true :-



(1) S_1 and S_2 must have the same intensities.

(2) S_1 and S_2 have a constant phase difference.

(3) S_1 and S_2 must have the equal phase.

(4) S_1 and S_2 must have the same wavelength.

30) A source of light is placed in front of a screen. Intensity of light on the screen is I. Two Polaroids P_1 and P_2 are so placed in between the source of light and screen that the intensity of light on screen is I/2. P_2 should be rotated by an angle of ..... (degrees) so that the intensity of light on the screen becomes $\frac{3I}{8}$.

(1) 45

(2) 30

(3) 60

(4) 90

31) In YDSE, intensity of light passes through both slits is equal. A 100% transparent thin film is placed infront of one of the slits. Now the intensity on the centre becomes 25% of the previous intensity. The wavelength of light is 4000 Å and refractive index 1.5. The minimum thickness of the glass slab is :-

(1) 1.23×10^{-7} m

(2) 2.67×10^{-7} m

(3) 4×10^{-7} m

(4) None of these

32) In YDSE the path difference between two interfering waves at a point on the screen is λ/6. The ratio of intensity at this point and that at the central bright fringe will be : (Assume that intensity due to each slit in same)

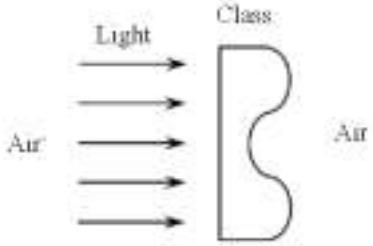
(1) 0.853

(2) 8.53

(3) 0.75

(4) 7.5

33) A parallel beam of light strikes a piece of transparent glass having cross section as shown in the figure below. Correct shape of the emergent wavefront will be (figures are schematic and not drawn to scale)-



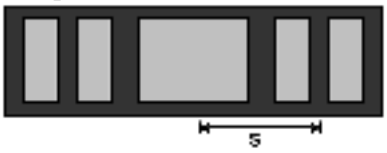
(1) 

(2) 

(3) 

(4) 

34) Light of wavelength 600 nm is incident upon a single slit with width 4×10^{-4} m. The figure shows the pattern observed on a screen positioned 2m from the slits. Determine the distance s.



(1) 0.002 m

(2) 0.003 m

(3) 0.004 m

(4) 0.006 m

35) The angular width of the central maximum in a single slit diffraction pattern is 60°. The width of the slit is 1 μm. The slit is illuminated by monochromatic plane waves. If another slit of same width is made near it, Young's fringes can be observed on a screen placed at a distance 50 cm from the slits. If the observed fringe width is 1 cm, what is slit separation distance? (i.e. distance between the centres of each slit.)

(1) 50 μm

(2) 75 μm

(3) 100 μm

(4) 25 μm

36) The electron mobility in N-type germanium is 3900 $\frac{cm^2}{V-s}$ and its conductivity is 6.24 $\frac{mho}{cm}$ then impurity concentration will be if the effect of hole is negligible :-

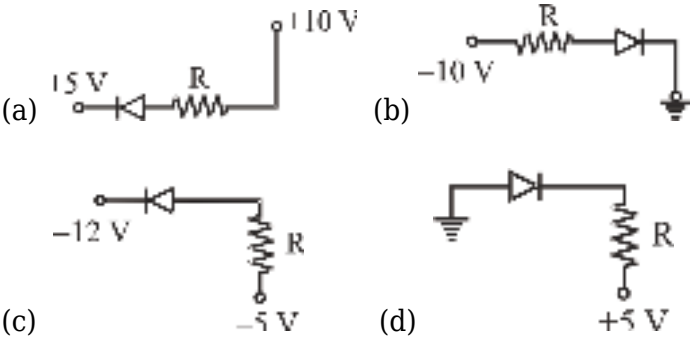
(1) $10^{15}/cm^3$

(2) $10^{13}/cm^3$

(3) $10^{16}/cm^3$

(4) $10^{16}/m^3$

37) In the following figure, the diodes which are forward biased, are :-



(1) (a), (b) and (d)

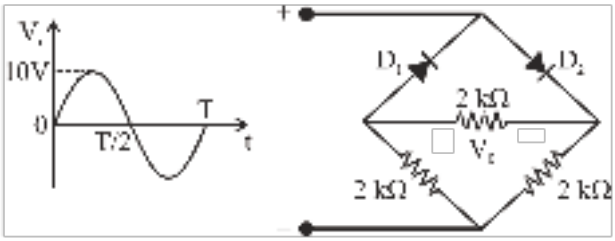
(2) (c) only

(3) (a) and (c)

(4) (b) and (d)

38)

In the circuit shown in figure the maximum output voltage V_0 is -



(1) 0 V

(2) 5 V

(3) 10 V

(4) $\frac{5}{\sqrt{2}}$ V

39) An AC supply of 100 V is applied to half wave rectifier through a transformer of turn ratio 2 : 1. Find output DC voltage :

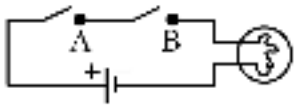
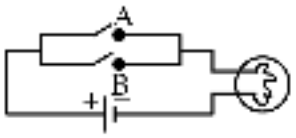
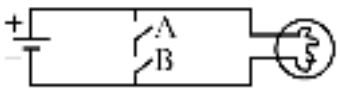
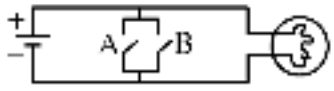
(1) $50\sqrt{2}$ V

(2) $\frac{50\sqrt{2}}{\pi}$ V

(3) $\pi 50\sqrt{2}$ V

(4) $\frac{\pi 50}{\sqrt{2}}$ V

40) Match the analog circuits given in column-I with their logical (operational) expression given in column-II.

Column-I		Column-II	
(a)		(p)	$\overline{\bar{A}+\bar{B}}$
(b)		(q)	$\overline{AB}$
(c)		(r)	$\overline{\bar{A}.\bar{B}}$
(d)		(s)	$\overline{A+B}$

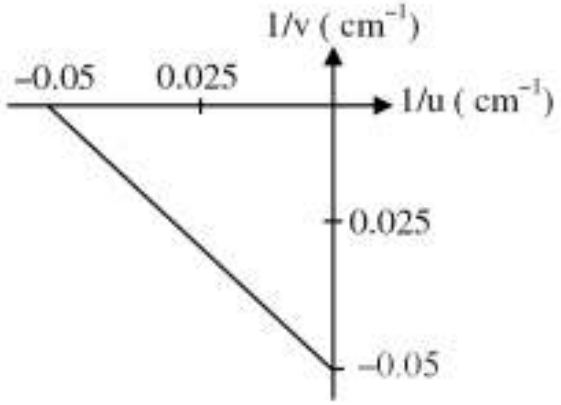
(1) a-p, b-q, c-r, d-s

(2) a-q, b-p, c-r, d-s

(3) a-p, b-r, c-q, d-s

(4) a-s, b-r, c-q, d-p

41) For concave mirror, a graph between 1/u versus 1/v is obtained using parallax method on optical bench as shown in figure. From graph, the estimated value of power of mirror is :-



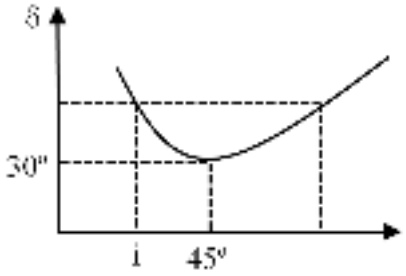
(1) +2D

(2) –2.5 D

(3) +4 D

(4) +5 D

42) A graph between angle of incidence and angle of deviation for a prism placed in air is as shown in figure. The refractive index for material of prism and critical angle, for given pair of medium is :-



(1) 1.5, 30°

(2) 1.25, 37°

(3) $\sqrt{2}$, 45°

(4) $\sqrt{2}$, 60°

43) A travelling microscope is used to measure refractive index of a glass slab and following observations are made :
a. Cross - mark without slab : 2.150 cm
b. Cross - mark with slab : 3.350 cm
c. Top surface with lycopodium powder: 5.350 cm
The calculated value of refractive index of slab is :

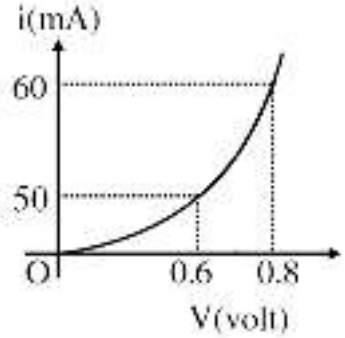
(1) 1.500

(2) 1.550

(3) 1.600

(4) 1.650

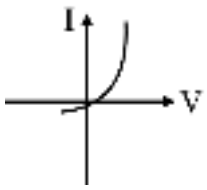
44) The forward characteristics of a p–n junction diode is as shown in figure, the dynamic resistance

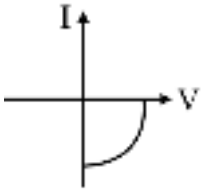


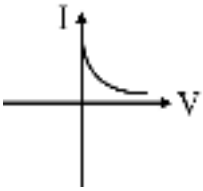
of diode at applied potential of 0.7 V is :

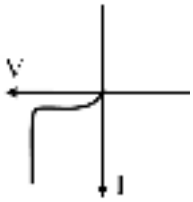
(1) 5 Ω
(2) 10 Ω
(3) 15 Ω
(4) 20 Ω

45) I-V characteristics of some semiconductor devices are as shown below. The correct order of devices in the respective order of their graph is :

(a) 

(b) 

(c) 

(d) 

(1) Zener diode, Simple diode, Solar cell, Capacitor
(2) Simple diode, Solar cell, Capacitor, Zener diode
(3) Zener diode, Capacitor, Solar cell, Simple diode
(4) Capacitor, Solar cell, Simple diode, Zener diode

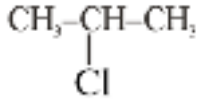
CHEMISTRY

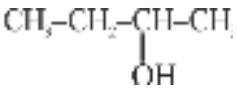
1) Consider thiol anion ($RS^{\ominus}$) and alkoxy anion ($RO^{\ominus}$). Which of the following statement is correct ?

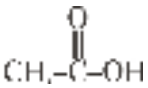
(1) $RS^{\ominus}$ is less basic but more nucleophilic than $RO^{\ominus}$
(2) $RS^{\ominus}$ is more basic and more nucleophilic than $RO^{\ominus}$
(3) $RS^{\ominus}$ is more basic but less nucleophilic than $RO^{\ominus}$
(4) $RS^{\ominus}$ is less basic and less nucleophilic than $RO^{\ominus}$

2) Which of the following compound can not give iodoform test :-

(1) $CH_3–CH_2–OH$

(2) 

(3) 

(4) 

3) Which is the most suitable reagent for the following conversion

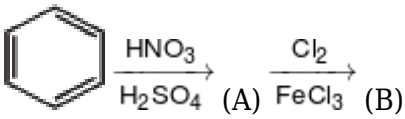


(1) $\xrightarrow[(ii)\ H^{\oplus}]{(i)\ \text{Tollen's reagent}}$

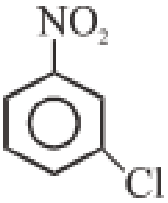
(2) $\xrightarrow[(ii)\ H^{\oplus}]{(i)\ \text{Benzoyl peroxide}}$

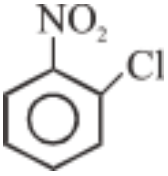
(3) $\xrightarrow[(ii)\ H^{\oplus}]{(i)\ I_2 \text{ and NaOH}}$

(4) $\xrightarrow{KMnO_4/H^{\oplus}, \Delta}$

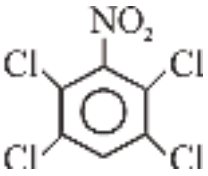
4) 

The major product (B) is :-

(1) 

(2) 

(3) 

(4) 

5) Which gives red colour with Fehling's solution ?

(1) Glucose

(2) Cellulose

(3) Benzaldehyde

(4) Cane sugar

6) The correct order of reactivity given carbonyl compounds towards nucleophilic addition reaction is :

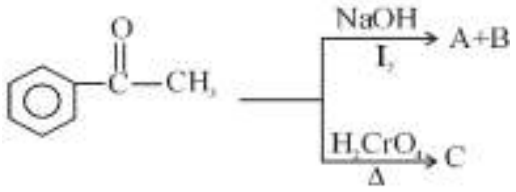
(1) Benzaldehyde > p-Tolualdehyde > p-Nitrobenzaldehyde

(2) p-Nitrobenzaldehyde > Benzaldehyde > p-Tolualdehyde

(3) p-Tolualdehyde > Benzaldehyde > p-Nitrobenzaldehyde

(4) None of these

7)



A, B and C respectively are-

(1) CHI_3, C_6H_5–COOH, C_6H_5–COONa

(2) C_6H_5–COONa, I_2, C_6H_5–CHO

(3) CHI_3, C_6H_5–COONa, C_6H_5–COOH

(4) CHI_3, C_6H_5–COONa, C_6H_5–CHO

8) Which of the following name reaction not involved in formation of aldehyde (–CHO) group :

(1) Etard reaction

(2) Gattermann – Koch reaction

(3) Clemensen's Reduction

(4) Rosenmund's Reduction

9) Aldehyde which do not have an α hydrogen atom, undergo self oxidation and reduction (disproportionation) reaction on heating with conc. alkali is called :

(1) Cannizzaro's reaction

(2) Aldol condensation reaction

(3) Hell-Volhard-Zelinsky reaction

(4) Claisen condensation reaction

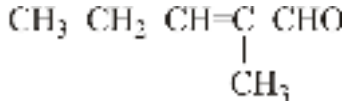
10) CH_3–CHO + CH_3–CH_2–CHO $\xrightarrow[2.\ \Delta]{1.\ NaOH}$ Products

Which of the following is not possible in this reaction :

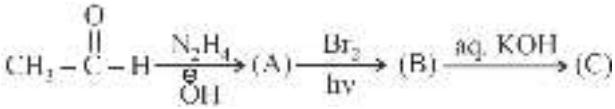
(1) CH_3–CH_2–CH=CH–CH_2–CHO

(2) CH_3–CH=CH–CHO

(3) CH_3–CH_2–CH=CH–CHO

(4) 

11) For the reaction :-



What is correct about product (C)?

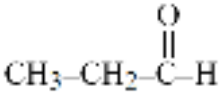
(1) (C) gives silver mirror with Tollen's reagent.

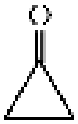
(2) (C) gives yellow precipitate with I_2/NaOH.

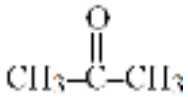
(3) (C) can also be prepared by the reaction of HCHO with methyl magnesium bromide followed by H^+

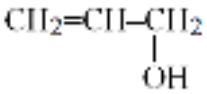
(4) Both (2) & (3) are correct.

12) An organic compound C_3H_6O forms an orange red precipitate with 2,4-D.N.P. but it does not give tollen's test compound is :-

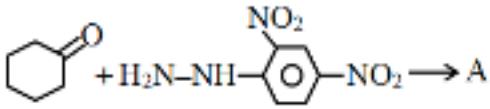
(1) 

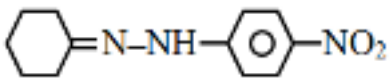
(2) 

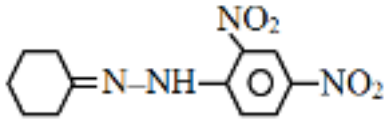
(3) 

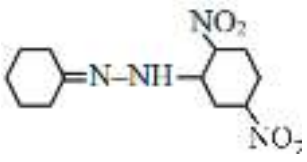
(4) 

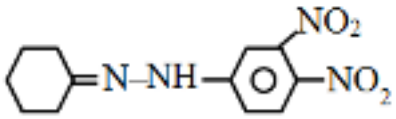
13) Identify product (A) in given reaction :-

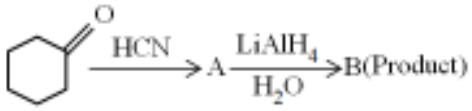


(1) 

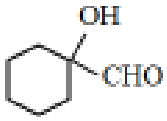
(2) 

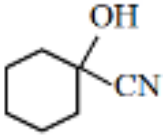
(3) 

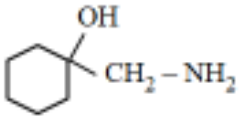
(4) 

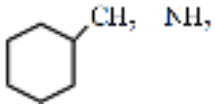
14) 

B will be :-

(1) 

(2) 

(3) 

(4) 

15) Methyl cyanide on treatment with methyl magnesium bromide followed by subsequent hydrolysis gives.

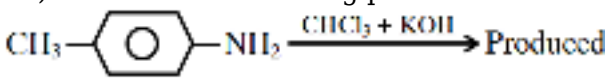
(1) Propanone

(2) Ethanone

(3) Ethanal

(4) Propanal

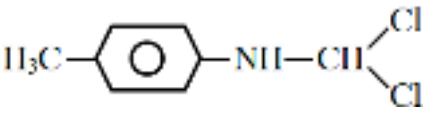
16) Which of the following products will form by following reaction?



(1) 

(2) 

(3) 

(4) 

17) **Statement-I :** Haloalkanes react with KCN to form alkyl cyanides as main product.
Statement-II : Haloalkanes react with AgCN to form alkyl isocyanides as the chief product.

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is correct, but statement-II is incorrect.

(4) Statement-I is incorrect, but Statement-II is correct.

18) Choose the correct reagents to bring out the following conversion.

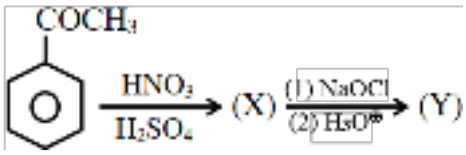
$\text{Benzene} \xrightarrow{?} 3-\text{Bromoaniline}$

(1) Br_2/Fe, conc HNO_3, Sn/HCl

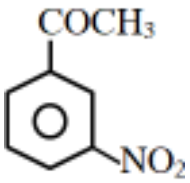
(2) Conc. HNO_3, Sn/HCl, Br_2/Fe

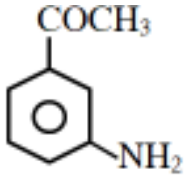
(3) Conc. HNO_3, Br_2/Fe, Sn/HCl

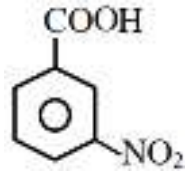
(4) Br_2/Fe, Sn/HCl, Conc. HNO_3

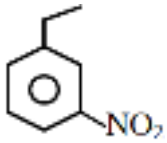
19) 

Identify of 'Y' :-

(1) 

(2) 

(3) 

(4) 

20) In the reaction sequence identify the functional groups present in A, B, C ?

$(A) \xrightarrow{Sn/HCl} (B) \xrightarrow{HNO_2, 0^0C} (C) \xrightarrow{C_2H_5OH} C_6H_6$ Choose the correct option :-

(1) A is $-NH_2$ group

(2) B is $-NH_2$ group

(3) C is –OH group

(4) B is $-NO_2$ group

21)

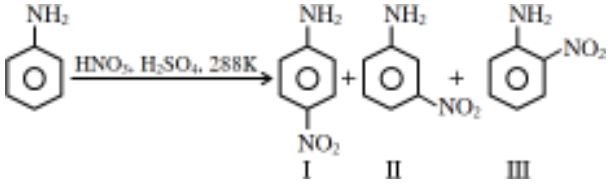
Which of the following statements is incorrect

(1) Secondary and tertiary amines show carbylamine reaction.

(2) Primary aliphatic amines react with nitrous acid to form aliphatic diazonium salts which being unstable, liberate nitrogen gas quantitatively and forms alcohol.

(3) Benzenesulphonyl choride is used for the distinction of primary, secondary and tertiary amines and also for the separation of a mixture of amines.

(4) N-methyl benzamide is formed when methanamine reacts with benzoyl chloride. This reaction is known as benzoylation.

22) 

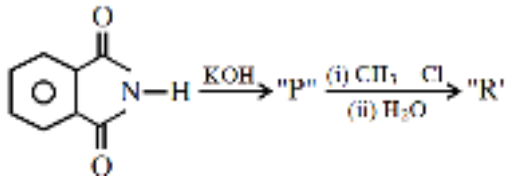
Select the percentage in which these amine derivatives are formed in the reaction mixture along with the reason.

(1) I → 2%, II → 47%, III → 51%
In the acidic medium, aniline is protonated to form the anilinium ion which is ortho directing

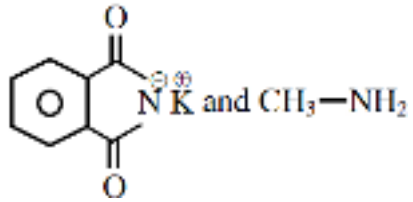
(2) I → 51%, II → 2%, III → 47%
In the acidic medium, aniline is protonated to form the anilinium ion which is para directing

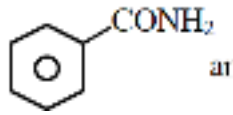
(3) I → 51%, II → 47%, III → 2%
In the acidic medium, aniline is protonated to form the anilinium ion which is meta directing

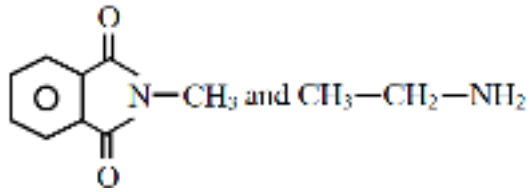
(4) I → 50%, II → 20%, III → 30%
In the acidic medium, aniline is protonated to form the anilinium ion which is ortho directing

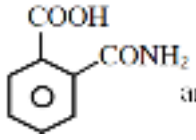
23) 

Here "P" and "R" are -

(1)  and CH_3-NH_2

(2)  and CH_3-NH_2

(3)  and $CH_3-CH_2-NH_2$

(4)  and R_2NH

24) $CH_3–CH_2–NH_2 + HNO_2 \rightarrow CH_2–CH_2–OH + 'x' + H_2O$
Here 'x' gas is-

(1) NH_3
(2) PH_3
(3) N_2
(4) HCl

25) Which among the following is non-reducing sugar?

(1) Glucose
(2) Maltose
(3) Lactose
(4) Sucrose

26) Which among the following statement is incorrect for glucose ?

(1) Glucose is an aldohexose
(2) Glucose shows mutarotation
(3) Glucose on reaction with HI/Δ give n-hexane
(4) α-D-glucose is monomer of cellulose

27) Vitamin C is chemically :-

(1) Ascorbic Acid
(2) Citric Acid
(3) Aspirin
(4) Aspartic Acid

28) Which of the following carbohydrate give same osazone ?

(1) D-Glucose, D-Galactose, D-Mannose
(2) D-Fructose, D-Galactose, D-Mannose
(3) D-Glucose, D-Mannose, D-Arabinose
(4) D-Glucose, D-Mannose, D-Fructose

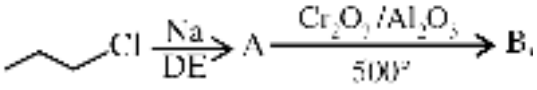
29) The hydrolysis product of (+) lactose is :-

(1) β–D(+)glucose + β–D(+)galactose
(2) α–D(+)galactose + α–D(+)glucose
(3) α–D(+)glucose + β–D(+)galactose

(4) β–D(+)glucose + α–D(+)glucose

30) In Duma's method for estimation of nitrogen, 0.25 g of an organic compound gave 40 mL of nitrogen collected at 300 K temperature and 725 mm pressure. If the aqueous tension at 300 K is 25 mm, the percentage of nitrogen in the compound is :-

(1) 18.20
(2) 16.76
(3) 15.76
(4) 17.36

31) 

B will be

(1) 
(2) 
(3) 
(4) 

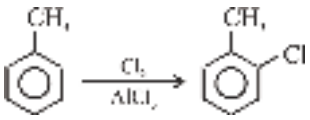
32) Paper chromatography is an example of:

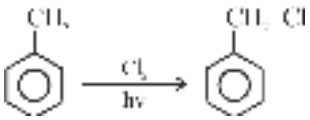
(1) Column chromatography
(2) Adsorption chromatography
(3) Partition chromatography
(4) Thin layer chromatography

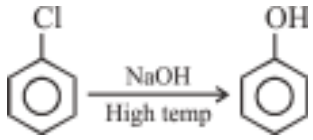
33) During Lassaigne's test for nitrogen, the prussian blue colour is obtained due to formation of

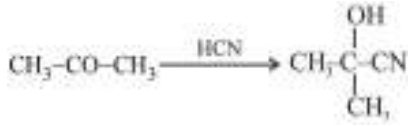
(1) $Na_4[Fe(CN)_6]$
(2) $Fe_4[Fe(CN)_6]_3$
(3) $Fe_2[Fe(CN)_6]$
(4) $Fe_3[Fe(CN)_6]_4$

34) Example of nucleophilic substitution reaction :-

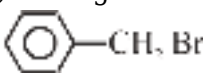
(1) 

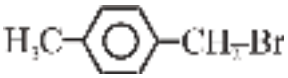
(2) 

(3) 

(4) 

35) Arrange in decreasing order towards hydrolysis of following compounds with S_N1 reactions :-

(I) 

(II) 

(III) 

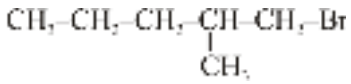
(IV) 

(1) III > IV > II > I

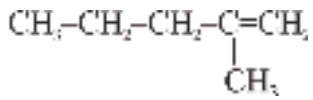
(2) IV > III > II > I

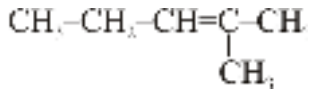
(3) II > III > IV > I

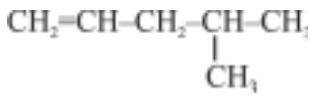
(4) I > II > III > IV

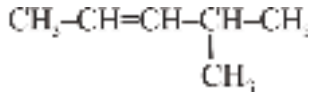
36) 

Which of the following products will be obtained if this alkyl halide is subjected to E2 reaction ?

(1) 

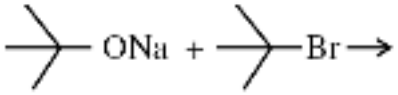
(2) 

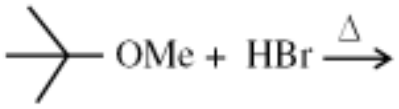
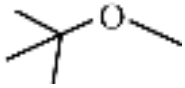
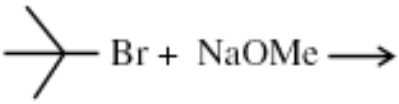
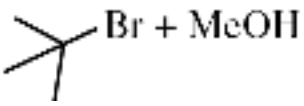
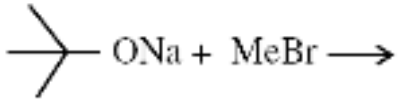
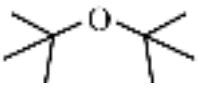
(3) 

(4) 

37)

Match column I with column II

Column-I (Reaction)		Column-II (Major Product)	
(A)		(P)	

(B)		(Q)	
(C)		(R)	
(D)		(S)	

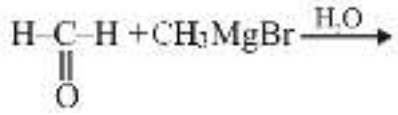
(1) A-S, B-R, C-P, D-Q

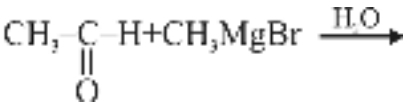
(2) A-S, B-P, C-R, D-Q

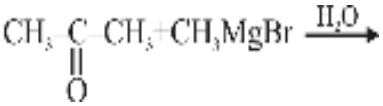
(3) A-P, B-Q, C-P, D-S

(4) A-P, B-R, C-P, D-Q

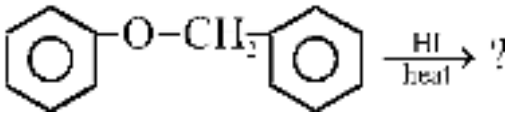
38) In which reaction 2° Alcohol is obtained :-

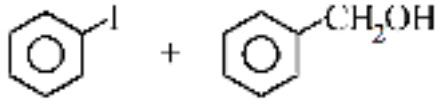
(1) 

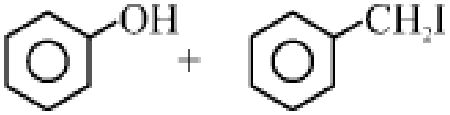
(2) 

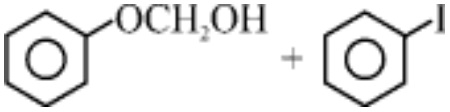
(3) 

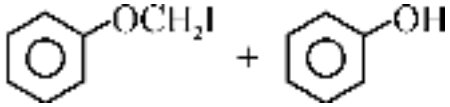
(4) None of these

39) What are the major products from the following reaction ? 

(1) 

(2) 

(3) 

(4) 

40) Best method to convert alcohol to halide is :-

(1) $ROH+SOCl_2 \rightarrow R\text{–}Cl+SO_2+HCl$

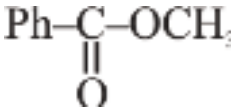
(2) $R\text{–}OH+PCl_3 \rightarrow R\text{–}Cl$

(3) $R–OH+PCl_5 \rightarrow R–Cl$

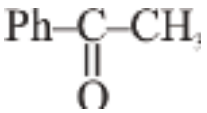
(4) $R–OH+HCl \rightarrow R–Cl+H_2O$

41) $PhMgBr \xrightarrow[(ii)\ H_2O]{(i)\ CO_2} A \xrightarrow{CH_3MgBr} B$ Product B is :

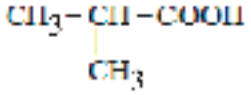
(1) CH_4

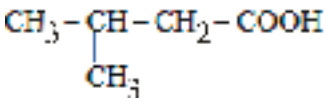
(2) 

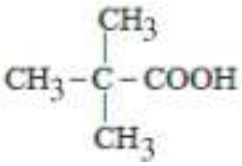
(3) $PhCH_2OH$

(4) 

42) Arrange these acids in decreasing order of ease of esterification with $CH_3OH/H^{\oplus}$:

(I) 

(II) 

(III) 

(IV) $(CH_3–CH_2)_3C–COOH$

(1) II > I > III > IV

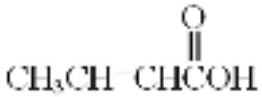
(2) I > II > III > IV

(3) III >IV > II > I

(4) IV > III > II > I

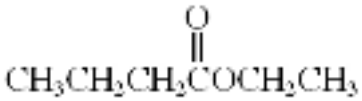
43) What is compound Z ?

$CH_3CH_2CH_2Br \xrightarrow{NaCN} X \xrightarrow[heat]{H_3O^+} Y \xrightarrow[H^+]{CH_3CH_2OH} Z$

(1) 

(2) $CH_3CH_2CH_2CH=NOCH_2CH_3$

(3) $CH_3CH_2CH_2CH(OCH_2CH_3)_2$

(4) 

44) In which of the following reaction acetic acid is formed?

(1) $CH_3–CN \xrightarrow{H_3O^{\oplus}}$

(2) $CH_3–CH_2–OH \xrightarrow[KMnO_4]{hot}$

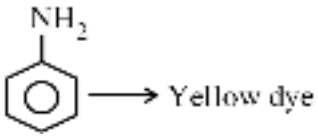
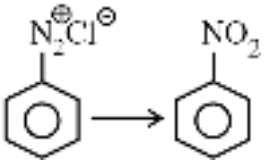
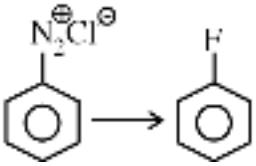
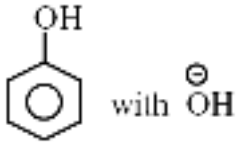
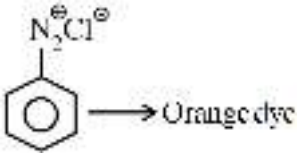
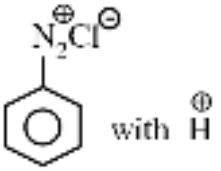
$CH_3–CH=CH–CH_3$

(3) $\xrightarrow[KMnO_4/H^{\oplus}]{Hot}$

(4) All of these

45)

Match column I with column II

Column-I (Reaction)		Column-II (Reagent)	
(P)		(i)	HBF_4 / Δ
(Q)		(ii)	HBF_4 followed by $NaNO_2$ / Cu, Δ
(R)		(iii)	
(S)		(iv)	

(1) P-(iii), Q-(ii), R-(i), S-(iv)

(2) P-(iii), Q-(i), R-(ii), S-(iv)

(3) P-(iv), Q-(ii), R-(i), S-(iii)

(4) P-(iv), Q-(i), R-(ii), S-(iii)

BIOLOGY

1) **Assertion (A) :** Net primary productivity (NPP) represents actual biomass available for herbivores and decomposers.
Reason (R) : NPP is calculated by subtracting respiration losses (R) from gross primary productivity (GPP).

(1) Both (A) and (R) are true and (R) is correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not correct explanation of (A).

(3) (A) is true, (R) is false

(4) Both (A) and (R) are false.

2) **(A) :** Ecosystems are not exempt from second law of thermodynamics. They need constant supply of energy to synthesize molecules they require, to counteract increasing disorderliness.
(B) : Plants capture only 2-10 percent of PAR.

(1) Both (A) and (B) are correct.

(2) Both (A) and (B) are incorrect.

(3) (A) is correct, (B) is incorrect.

(4) (A) is incorrect, (B) is correct

3)

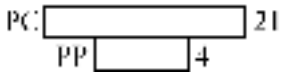
	Column-I		Column-II		Column-III
(1)	Producer	(i)	Second trophic level	(a)	Man
(2)	Primary consumer	(ii)	Third trophic level	(b)	Phytoplankton
(3)	Secondary consumer	(iii)	Fourth trophic level	(c)	Birds
(4)	Tertiary consumer	(iv)	First trophic level	(d)	Zooplankton

(1) 1-iii-b, 2-i-c, 3-iv-a, 4-ii-d

(2) 1-iii-a, 2-ii-b, 3-iv-d, 4-i-c

(3) 1-iv-a, 2-ii-d, 3-iii-b, 4-i-c

(4) 1-iv-b, 2-i-d, 3-ii-c, 4-iii-a

4) An ecological pyramid is given below :- 

(PP stands for primary producer), (PC stands for primary consumer)
Given ecological pyramid represents :-

(1) Pyramid of number in a tree ecosystem

(2) Pyramid of biomass in aquatic ecosystem

(3) Pyramid of energy in grassland ecosystem

(4) Pyramid of number in grassland ecosystem

5) Read the following statements and select correct ones :-
a. Herbivores are also called secondary producers.
b. Bacteria and Fungi are microconsumers.
c. Rabbit, deer, goat are secondary producers
d. Mode of nutrition of macroconsumer is first ingest then digest

(1) only a, b, c

(2) only b, c, d

(3) only a, c, d

(4) a, b, c, d

6) Read the following statements and select correct one :-

(1) Gross primary productivity is the available biomass for the consumption to heterotrophs,

(2) Secondary productivity depends on the photosynthetic capacity of the plant species inhabiting a particular area.

(3) Productivity is expressed in terms of g m^{-2} or kcal m^{-2} to compare the productivity of different ecosystems.

(4) The annual NPP of the whole biosphere is 170 billion tons of organic matter.

7) Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): Pyramid of energy is always upright, can never be inverted.
Reason (R): When energy flows from a particular trophic level to the next trophic level, some energy is always lost as heat at each step.
In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

8) How many aspects in the list given below are functional aspects of an ecosystem ?
List :- Stratification, Decomposition, nutrient cycling, species composition, physical structure, productivity.
Choose the correct answer from the options given below.

(1) Three

(2) Four

(3) One

(4) Five

9) What would be the consequence if humification is impaired in a terrestrial ecosystem?

(1) Rapid nutrient turnover

(2) Accumulation of nutrients decreases

(3) Increase in soil fertility

(4) Excessive fragmentation

10) What is the most likely effect of reduced species diversity in a tropical forests ?

(1) Increased food web complexity

(2) Higher resilience to invasion

(3) Reduced ecosystem productivity

(4) Enhanced carbon fixation

11) Which of the following is **correct** regarding decomposition ?

(1) Catabolism breaks down the detritus into smaller particles
(2) Mineralisation produces dark amorphous material
(3) Leaching removes soluble substances from top soil
(4) Humification is faster in dry, low microbial environments

12) Match the Columns

	Column I (Interaction)		**Column II (Example)**		**Column III (Effect)**		**Column IV (Presentation)**
(A)	Commensalism	(1)	Cattle egret + cow	(i)	Only one benefited	α	+/0
(B)	Parasitism	(2)	Cuscuta on hedge	(ii)	Host exploited	β	–/+
(C)	Mutualism	(3)	Mycorrhizae	(iii)	Both benefited	γ	+/+
(D)	Predation	(4)	Snake eating frog	(iv)	One kills other	δ	+/–

(1) A–1–ii–γ; B–3–i–α; C–2–iii–β; D–4–iv–δ
(2) A–2–ii–β; B–1–iii–γ; C–3–i–α; D–4–iv–δ
(3) A–3–iii–γ; B–4–iv–α; C–2–ii–β; D–1–i–δ
(4) A–1–i–α; B–2–ii–β; C–3–iii–γ; D–4–iv–δ

13) Which of the following ecological pyramid is always upright ?

(1) Pyramid of number in a forest
(2) Pyramid of number in a tree ecosystem
(3) Pyramid of energy in a pond
(4) Pyramid of biomass in aquatic ecosystem

14) Given below are two statements :
Statement I :- A constant amount of solar energy is the basic requirement for any ecosystem to sustain and for functionality.
Statement II :- Primary production is defined as the amount of biomass or organic matter produced per unit area over a time by plants during photosynthesis.
In the light of above statements, choose the correct answer from the options given below :

(1) Both **Statement I** and **Statement II** are correct.
(2) Both **Statement I** and **Statement II** are incorrect.
(3) **Statement I** is correct but **Statement II** is incorrect.
(4) **Statement I** is incorrect but **Statement II** is correct.

15) Which of the following is **not** correct regarding competition ?

(1) Always competitive exclusion
(2) Co existence through niche partitioning

(3) Competition between unrelated species

(4) Competition even if resources are abundant

16) Choose the **correct** statements :-
(I) Pollination by animals is a mutualistic interaction.
(II) Generally predators eliminate the prey species completely (extinct) in nature.
(III) In mutualism only one partner get benefited
(IV) Predators are harmful for ecosystem stability.
Options :-

(1) Only IV

(2) Only II, III

(3) Only I

(4) Only I and II

17) Given below are two statements; one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): Predators help in maintaining species diversity in a community. **Reason (R):** Predators reduce the intensity of competition among competing prey species.
In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

18) Given below are two statements:
Statement-I: A wide variety of chemical substances that we extract from plants on a commercial scale (nicotine, caffeine, quinine, strychnine, opium, etc.,) are produced by them actually as defences against grazers and browsers.
Statement-II: Plants have evolved an astonishing variety of morphological and chemical defences against herbivores. Thorns (*Acacia*, Cactus) are the most common morphological means of defence.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

19) The interaction in which one species benefits and the other is neither harmed nor benefited is called _____**(A)**_____. An example of this is the relationship between the _____**(B)**_____ and the grazing cattle, where one species benefits by catching insects stirred up by the other.

(1) **A** = Parasitism, **B** = Tick

(2) **A** = Mutualism, **B** = Clown fish

(3) **A** = Commensalism, **B** = Egret

(4) **A** = Competition, **B** = Whale

20)

Which of the following statement is **correct** about species that significantly affects the balance of ecosystems?

(1) Alien species naturally evolve in a particular ecosystem and always promote biodiversity.

(2) Endemic species are dominant in every habitat where they are found and typically determine the biodiversity.

(3) Endemic species are found in a particular area so their chances of extinction are very less

(4) Keystone species may be few in number, but their removal often leads to drastic changes in the ecosystem, as they help regulate population sizes and species interactions.

21) Which of the following factor can reduce the intensity of competition ?

(1) Limited resources

(2) Resource partitioning

(3) Close relation between species

(4) Increase in population density

22) **Assertion (A) :-** Interspecific competition can sometimes lead to the extinction of one of the competing species.
Reason (R) :- According to Gause's competitive exclusion principle, two closely related species competing for the same resources can coexist indefinitely in a stable environment.

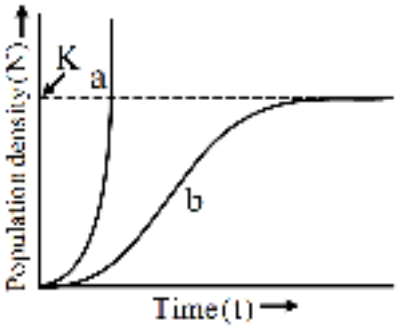
(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) Both A and R are false

23) Select the **correct** option with respect to given population growth curves :



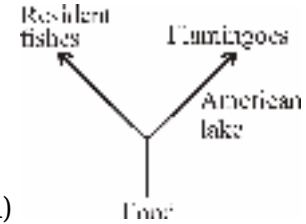
(1) **a** = resources are not limiting the growth; $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$

(2) **b** = resources are limiting the growth; $\frac{dN}{dt} = rN$

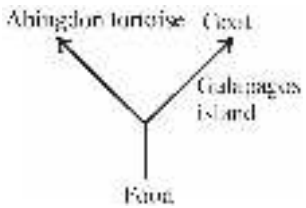
(3) **a** = growth is exponential; $N_t = N_0e^{rt}$

(4) **b** = growth is logistic ; $\frac{dN}{dt} rN = \left(\frac{N-K}{K}\right)$

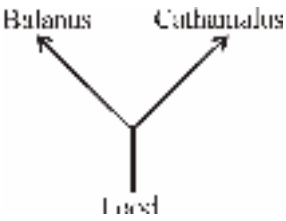
24) (A)



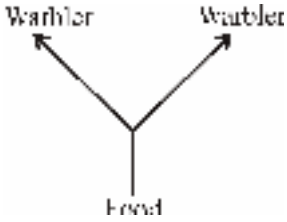
(B)



(C)



(D)



Choose the **incorrect** match :-

(1) **A** – Representing competition

(2) **B** – Showing competitive exclusion

(3) **C** – showing parasitism

(4) **D** – showing resources partitioning

25) Match the items in column-I with those in column-II

Column-I		**Column-II**	
(a)	Competition	(i)	Lichens
(b)	Predation	(ii)	Orchid-Mango
(c)	Commensalism	(iii)	Sparrow eating a seed
(d)	Mutualism	(iv)	Sheep-Deer

Select the correct option from following :

(1) a- iv, b - iii, c - ii, d - i

(2) a- iv, b - iii, c - i, d - ii

(3) a- iii, b - iv, c - ii, d - i

(4) a- i, b - ii, c - iii, d - iv

26) Match the column-I, II and III and select the correct one :-

	Column-I		**Column-II**		**Column-III**
a.	+, +	(i)	Commensalism	P.	Fig tree and wasp
b.	+, 0	(ii)	Mutualism	Q.	Whale and Barnacles
c.	Standing state	(iii)	Breakdown detritus in to smaller particles	R.	C, N, P

d.	Detritivore	(iv)	amount of inorganic nutrient	S.	Earthworm

(1) a-(i)-Q, b-(ii)-P, c-(iv)-R, d-(iii)-S

(2) a-(i)-Q, b-(ii)-P, c-(iii)-S, d-(iv)-R

(3) a-(ii)-P, b-(i)-Q, c-(iv)-R, d-(iii)-S

(4) a-(ii)-P, b-(i)-Q, c-(iii)-R, d-(iv)-S

27) Refer the given table that summarizes the interactions between two organisms (organism 1 and organism 2). Identify the types of interaction (A, B and C) and select the correct answer.

	Effects on Organism 2			
Effects on Organism 1		Benefit	Harm	No Effect
	Benefit	Mutualism	Predation	B
	Harm	A	D	Amensalism
	No Effect	Commensalism	C	–

(i) A can be either Predation or Amensalism.
(ii) B can be Commensalism.
(iii) C can be Amensalism.
(iv) D can be Parasitism.

(1) (i) and (ii), (iv)

(2) (ii) and (iii)

(3) (iii) and (iv)

(4) (i) and (iii), (iv)

28) In the species-area curve, if Z = 1.2, this indicates :-

(1) A very low rate of species accumulation

(2) A typical pattern in small areas

(3) Greater species richness in larger areas

(4) Constant number of species per unit area

29) Which of the following examples best represents species diversity ?

(1) Over 50,000 varieties of rice grown in India

(2) Diverse amphibians in Western Ghats

(3) Different types of ecosystems in India

(4) Genetic variations among Himalayan wolves

30) Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason(R)**.
Assertion (A): Rich biodiversity is not only essential for ecosystem health but imperative for the very survival of the human race on this planet.
Reason (R): Tilman found that plots with more species showed less year-to-year variation in total biomass. He also showed that in his experiments, increased diversity contributed to higher productivity.
In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

31) The diversity of plants and animals is not uniform throughout the world but shows a rather uneven distribution. For many group of animals or plants, there are interesting patterns in diversity, the most well - known being the latitudinal gradient in diversity. Which of the following statements about the Amazon rainforest is **correct**?

(1) It is located in a polar region

(2) It supports fewer species than temperate forests.

(3) It has the greatest biodiversity on Earth.

(4) It has only a few insect species.

32) Which of the following is an advanced ex-situ conservation?

(1) Botanical garden

(2) Zoological park

(3) Seed Bank

(4) Wildlife safari park

33) **Assertion (A):** Tropical latitudes have greater biological diversity then temperate latitude.
Reason (R): Tropical regions remain relatively undisturbed for million of years.

(1) Both (A) and (R) are true and (R) is correct explanation of (A)

(2) Both (A) and (R) are true but R is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

34) India is one of the twelve megadiversity countries with _____ share in the global species diversity.

(1) 12.1 %

(2) 18.1 %

(3) 38.1%

(4) 8.1 %

35) Which of the following is **not** the part of "The Evil Quartet" causing biodiversity loss ?

(1) Habitat loss and fragmentation

(2) Over exploitation

(3) Allen species invasions

(4) Climate change

36) Which of the following is **wrong** ?

(1) Mammals are less diverse than fishes
(2) Amphibians are less diverse than birds
(3) Birds are less diverse than reptiles
(4) Fishes are more diverse than birds

37) Which one of the following is **not** an example of alien species in India?

(1) *Parthenium*
(2) *Clarias gariepinus*
(3) Nile perch
(4) *Lantana*

38) The hot spots of biodiversity conservation are characterized by :-

(1) High endemicity and high threat of extinction
(2) Low endemicity and high threat of extinction
(3) High endemicity and low threat of extinction
(4) Low endemicity and low threat of extinction

39) Which one of the following is a broadly utilitarian aspect of conservation of biodiversity ?

(1) Source of O_2
(2) Food
(3) Medicines
(4) Industrial products

40) Extinction of Steller's sea cow and passenger pigeon was due to-

(1) Habitat loss and fragmentation
(2) Co-extinction
(3) Alien species invasion
(4) Over exploitation

41) **Assertion (A) :-** Carrying capacity is the maximum population size a habitat can support.
Reason (R) :- Beyond carrying capacity, no further growth of population is possible.

(1) Both A and R are true and R explains A
(2) Both A and R are true but R doesn’t explain A
(3) A is true; R is false
(4) A is false; R is true

42) A population of butterflies has 800 individuals. During summer, 120 individuals die, 150 are born, 60 immigrate, and 40 emigrate. What is the net population size after summer?

(1) 850
(2) 830

(3) 790

(4) 780

43) What characteristic is shared by predation, parasitism and commensalism ?

(1) One species always benefits while the other is harmed

(2) The interacting species live closely together

(3) They always lead to extinction of one species

(4) They involve cooperation between both the species

44) Which of the following is the most appropriate measure of population density for a single large banyan tree in a field of many small carrot grass plants ?

(1) Number of banyan seeds

(2) Number of leaves on tree

(3) Percent cover or biomass

(4) Number of tree branches

45) Match **List-I** with **List-II**

	List-I		**List-II**
(A)	Population size (N) of any species is	(I)	$\frac{dN}{dt} = rN$
(B)	Increase or decrease in N during unit time period 't'	(II)	Not a static parameter
(C)	Verhulst pearl logistic growth	(III)	$\frac{dN}{dt} = rN\frac{(K-N)}{K}$
(D)	Integral form of exponential growth equation	(IV)	$Nt = N_0e^{rt}$

Choose the correct answer from the options given below :

(1) A-II, B-III, C-I, D-IV

(2) A-II, B-IV, C-I, D-III

(3) A-II, B-IV, C-III, D-I

(4) A-II, B-I, C-III, D-IV

46) **Statement-I :** Passive immunity provides immediate protection to an individual but it is short lived and does not produce immunological memory against future infections.
Statement-II : Active immunity develops more slowly but is long lasting because the body generates both plasma cells producing antibodies and memory cells. Which are capable of rapid response.

(1) Both statement-I and statement-II are true

(2) Both statement-I and statement-II are false

(3) Only statement-I is true

(4) Only statement-II is true

47) Read the following paragraph and answer the question -
The immune system defends the body against pathogens. The cells involved in immune responses include B-Lymphocytes, which produce antibodies and T-Lymphocytes which help in cell mediated immunity vaccine work on the principle of stimulating the immune system to recognise and respond quickly to pathogens during future infections.
Statement-I : Vaccine provide passive immunity to the body by introducing preformed antibodies.
Statement-II : The immunity developed after vaccination lasts for long durations due to the formation of memory cells.

(1) Both Statement-I and Statement-II are correct.

(2) Statement-I is correct and Statement-II is incorrect.

(3) Both Statement-I and Statement-II are incorrect.

(4) Statement-I is incorrect and Statement-II is correct.

48) How many of the following are bacterial diseases ?
(i) Typhoid
(ii) Common cold
(iii) Pneumonia
(iv) Dysentery
(v) Malaria
(vi) Ascariasis
(vii) Alkaptonuria

(1) Two

(2) Three

(3) Four

(4) Five

49) **Assertion (A) :** Acquired immunodeficiency syndrome is not classified as congenital disease because it is not inherited genetically from the parents and does not occur due to any chromosomal abnormality present at birth.
Reason (R) : AIDS is transmitted through sexual contact, blood transfusion and contaminated needles.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true and R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

50) Which of the following options gives the correct matching of a disease with its causative organisms and mode of transmission?

	Disease		**Causative organism**		**Mode of transmission**
(i)	Pneumonia	(a)	Microsporum	(p)	Intense itching
(ii)	Typhoid	(b)	Plasmodium	(q)	Fluid filled alveoli

(iii)	Ring worm	(c)	Strepto coccus pneumoniae	(r)	Intestinal perforation
(iv)	Malaria	(d)	Salmonella typhi	(s)	Fever with chill

(1) (i) - (a) - (p), (ii) - (b)-(q), (iii)-(c)-(r), (iv)-(d)-(s)

(2) (i) - (b) - (q), (ii) - (a)-(r), (iii)-(c)-(s), (iv)-(d)-(p)

(3) (i) - (c) - (q), (ii) - (d)-(r), (iii)-(a)-(p), (iv)-(b)-(s)

(4) (i) - (c) - (q), (ii) - (a)-(r), (iii)-(b)-(p), (iv)-(d)-(s)

51) Read the following statement (A)-(D) and choose the True (T) and False (F).
(A) Non-ionizing radiations like X-rays and gamma rays cause DNA damage leading to meoplastic transformation.
(B) Antibodies against cancer specific antigens are used for detection of certain cancer.
(C) α-interferen is a biological response modifier
(D) Metastasis is the most feared property of benign tumors.

	(A)	**(B)**	**(C)**	**(D)**
(1)	(F)	(T)	(F)	(F)
(2)	(T)	(T)	(F)	(F)
(3)	(T)	(F)	(F)	(F)
(4)	(F)	(T)	(T)	(F)

(1) 1

(2) 2

(3) 3

(4) 4

52)

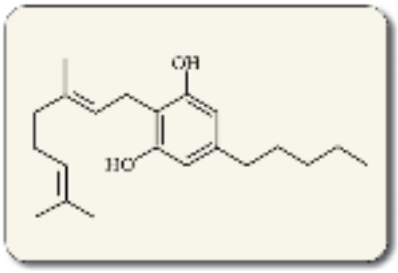
Read the following
a. Interferon
b. Skin
c. PMNLs
d. Mucus coat
e. Acid in stomach
f. Tears from eye
How many of the given barriers are not of physiological barrier type?

(1) 2

(2) 4

(3) 3

(4) 5

53) 

Choose the correct option for the given diagram of drug.

(1) Extracted from the latex of poppy papaver somniferum

(2) Obtained from the inflorescences of the plant cannabis sativa.

(3) It interferes with the transport of the neuro-transmitter dopamine.

(4) Used as medicines to help patient cope with physical illness.

54) Given below are two statements-
Statement-I : The secondary lymphoid organs provide the sites for interaction of lymphocytes with antigen, which then proliferate to become effector cells.
Statement-II : Both the bone marrow and thymus provide micro-environment for the development and maturation of B-lymphocytes.
In the light of the above statements select the correct answer from the options given below-

(1) Statement-I is incorrect but statement-II is correct.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I and statement-II both are incorrect.

(4) Statement-I and statement-II both are correct.

55) Given below are two statements :-
Statement-I : Entamoeba histolytica is a protozoan parasite in the small intestine of human.
Statemen-II : Ringworm are generally acquired from soil or by using towels, clothes or even the comb of infected individuals.
Choose the correct answer from the options given below :-

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

56) **Assertion (A) :** Pneumonia is a bacterial disease caused by streptococcus pneumoniae
Reason (R) : In severe cases of pneumonia, the lips and finger nails may turn grey to bluish in colour.

(1) If both assertion and reason are true and reason is correct explanation of assertion.

(2) If both assertion and reason are true but reason is not correct explanation of assertion.

(3) If assertion is true but reason is false.

(4) If both assertion and reason are false.

57) Match list-I with list-II :-

	List-I		List-II
A.	Morphine	I.	Claviceps purpurea
B.	Charas	II.	Erythroxylum coca
C.	Cocaine	III.	Cannabis sativa
D.	LSD	IV.	Papaver somniferum

Choose the correct option :-

(1) A-III, B-IV, C-II, D-I

(2) A-III, B-IV, C-I, D-II

(3) A-IV, B-III, C-I, D-II

(4) A-IV, B-III, C-II, D-I

58) Match the column-I with column-II and choose the correct match option :-

	Column-I		Column-II
A.	Colostrum	(i)	Active acquired
B.	Vaccination	(ii)	Passive acquired
C.	Natural killer cell	(iii)	Innate cellular barrier
D.	Interferon	(iv)	Innate cytokine barrier

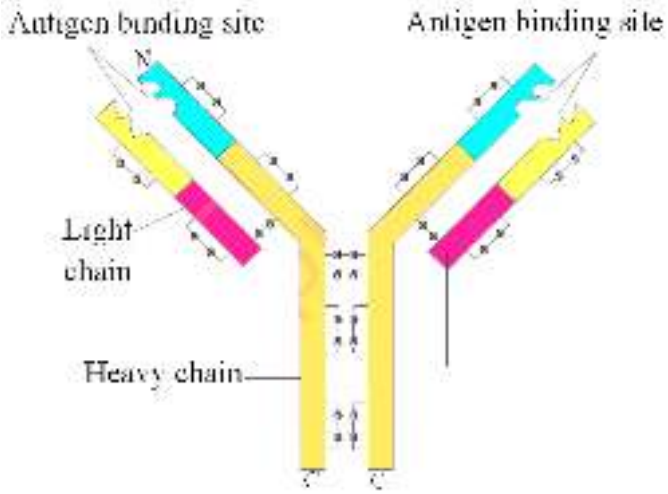
(1) A-(ii), B-(i), C-(iii), D-(iv)

(2) A-(ii), B-(i), C-(iv), D-(iii)

(3) A-(iv), B-(iii), C-(ii), D-(i)

(4) A-(iii), B-(ii), C-(i), D-(iv)

59) Which of the following statement is false for the structure given below ?



(1) Represented by H_2L_2

(2) These are found in blood or humor so response is termed as humoral immune response

(3) These are secreted by T cells

(4) They have four peptide chain

60) Which of the following statement is incorrect?

(1) Well-known plants with hallucinogenic properties are Atropa belladona and Datura.
(2) Tobacco chewing is associated with increased risk of cancer of the lungs.
(3) Withdrawal syndrome is characterised by anxiety, shakiness, nausea and sweating.
(4) The side-effects of the use of anabolic steroids in females include enlargement of clitoris and deepening of voice.

61) Find out the incorrect statement regarding lymphoid organs :-

(1) The primary lymphoid organs are bone marrow and thymus
(2) In primary lymphoid organs immature lymphocyte differentiate into antigen-sensitive lymphocyte.
(3) The primary lymphoid organs provide the site for interaction of lymphocytes with the antigen.
(4) The thymus is a lobed organ located near the heart and beneath the breast bone.

62) Good humor hypothesis of health was disproved by

(1) William Harvey
(2) Edward Jenner
(3) Alexander Fleming
(4) Hippocrates

63) Nicotine raise blood pressure and increase heart rate due to release of :-

(1) Cortisol
(2) Adrenaline
(3) Nor-Adrenaline
(4) More than one option is correct

64) Adolescence is a very vulnerable phase of mental and psychological development of an individual because

(1) The brain is fully matured, leading to overconfidence and recklessness
(2) Adolescence is accompanied by several biological and behavioural changes
(3) Peer pressure and emotional instability are at their minimum
(4) Adolescents face no major biological or hormonal changes

65) Which of the following drugs which are normally used as medicines to help patients cope with mental illness like depression and insomnia are often abused ?

(1) Barbiturates
(2) Amphetamines
(3) Benzodiazepines
(4) All of the above

66) Which of the following is given to the cancer patient to activate their immune system and to help

in destroying the tumor?

(1) Chemotherapy
(2) Radiotherapy
(3) Gamma-interferon
(4) Alpha-interferon

67) In the given symptoms which symptom is not associated with AIDS ?

(1) Weight loss
(2) Diarrhoea
(3) Bouts of fever
(4) Swelling in lower limbs

68) Gambusia is a fish which is being introduced into the ponds in order to check the vector borne disease such as :-

(1) Chikungunya
(2) Malaria
(3) Ascariasis
(4) Both (1) & (2)

69) Which is related to rhino virus ?

(1) Common cold
(2) Infection of nasal & respiratory passage
(3) Sore throat
(4) All of the above

70) Which of the following drug quickly reduce the symptoms of allergy?

(1) Anti histaminic
(2) Steroids
(3) Adrenaline
(4) All of these

71) Sometimes, due to genetic and other unknown reasons, the body attacks _____ called autoimmunity.

(1) Foreign cells
(2) Transplant cells
(3) Bacterial cell
(4) Self cells

72) **Statement-I :** In South Africa, mammals resembling horse, hippopotamus, bear, rabbit etc. were present. Due to continental drift, when South Africa joined North America, these animals were over ridden by North American Fauna.

Statement-II : Ernst Heckel noted that embryos never pass through the adult stage of other animals.

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct and statement-II incorrect

(4) Statement-I is incorrect and statement-II correct

73) Read the following statements and choose the correct option.
(A) Natural selection is based on certain observations which are hypothetical.
(B) Microbes that divide fast have the ability to multiply and become millions of individuals within hours.
(C) Some organisms are better adapted to survive in an otherwise hostile environment.
(D) Darwin observed that populations are stable in size except for seasonal fluctuation.

	(A)	(B)	(C)	(D)
(1)	F	T	T	T
(2)	F	T	T	F
(3)	T	F	T	T
(4)	F	T	F	T

(1) 1

(2) 2

(3) 3

(4) 4

74) How many of the following represent convergent evolution ?
(a) Bobcat - Tasmanian tiger cat
(b) Spotted cuscus - Lemur
(c) Sugar glider - Flying phalanger
(d) Anteater - Numbat
(e) Wombat - Kangaroo

(1) Three

(2) Four

(3) Two

(4) Five

75) Study the following events and arrange them in correct order
(A) Hydrogen and Helium formed
(B) A singular huge explosion
(C) Temperature came down
(D) Gases condensed under gravitation and formed the galaxies
(E) Universe expanded
Choose the correct option

(1) A, C, D, E, B

(2) B, C, E, A, D

(3) B, E, C, A, D

(4) A, D, B, E, C

76) Read the following terminology of column-I and II and identify their correct match :-

Column-I		Column-II	
(A)	The first non cellular forms of life	(i)	2-billion years ago
(B)	The first cellular form of life	(ii)	About 4 bya
(C)	Life appeared	(iii)	Originated 4.5 bya
(D)	Universe origin	(iv)	Protobionts
		(v)	3-billion years ago
		(vi)	About 13.8 bya

(1) (A)-(v), (B)-(i), (C)-(ii), (D)-(iii)

(2) (A)-(iv), (B)-(i), (C)-(iii), (D)-(vi)

(3) (A)-(iv), (B)-(ii), (C)-(v), (D)-(iii)

(4) (A)-(v), (B)-(i), (C)-(ii), (D)-(vi)

77) Read the following Assertion (A) and Reason (R) and choose the correct option
Assertion (A): Spontaneous generation theory was dismissed by Louis Pasteur.
Reason (R): In swan neck flask experiment, in pre-sterilised flasks, life did not come from killed yeast.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is incorrect

(4) Both (A) and (R) are incorrect

78) Read the following Assertion (A) and Reason (R) and choose the correct option
Assertion (A): In a mixed population, those that can better-adapt, survive and increase in population size.
Reason (R): Intensive breeding programme has created breeds that differ from other breeds but still are of the same group.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is incorrect

(4) Both (A) and (R) are incorrect

79) Given below are two statements :-
Statement-I : The sum of all genotype frequencies under Hardy-Weinberg equilibrium is 1.
Statement-II : In Hardy-Weinberg equilibrium, the allele frequencies remain constant from generation after generation if random mating does not occur.
In the light of the above statements, choose the most appropriate answer from the options given

below :-

(1) Both statement-I and statement-II are correct.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but statement-II is correct.
(4) Both statement-I and statement-II are incorrect.

80) Match the Column I with Column II and choose the correct option

	Column-I		**Column-II**
(A)	Triceratops	(i)	Flesh eating dinosaurs
(B)	Stegosaurus	(ii)	3 horned dinosaurs
(C)	Tyrannosaurus	(iii)	Missing link
(D)	Archeopteryx	(iv)	Spikes on Tail

	(A)	**(B)**	**(C)**	**(D)**
(1)	(i)	(ii)	(iii)	(iv)
(2)	(ii)	(iii)	(i)	(iv)
(3)	(ii)	(iv)	(i)	(iii)
(4)	(i)	(iii)	(iv)	(ii)

(1) 1
(2) 2
(3) 3
(4) 4

81) Match the column-I with column-II

	Column-I		**Column-II**
(a)	Jawless fishes evolved	(i)	350 mya
(b)	Sea weeds existed	(ii)	65 mya
(c)	Dinosaurs disappeared	(iii)	200 mya
(d)	Fish like reptiles evolved	(iv)	320 mya

Select the correct options :-

	(a)	(b)	(c)	(d)
(1)	(i)	(ii)	(iii)	(iv)
(2)	(iv)	(ii)	(i)	(iii)
(3)	(i)	(iv)	(ii)	(iii)
(4)	(iii)	(iv)	(i)	(ii)

(1) 1

(2) 2

(3) 3

(4) 4

82) Which of the following statement is incorrect?

(1) Existing living forms share similarities to varying degrees not only among themselves but also with life forms that existed millions of years ago.

(2) There had been extinctions of different life forms in the years gone by just as new forms of life arose at different periods of history of earth.

(3) All the existing life forms share similarities but do not share common ancestors.

(4) Diversity of life forms on earth has been changing over millions of years.

83) Among the following choose the incorrect statement.

(1) Evolutionary biology is the study of history of life forms in universe.

(2) To understand changes in flora and fauna we must understand context of origin of life.

(3) When we look at stars on clear night sky we are in way, looking back in time.

(4) When we see stars we are peeping into past.

84) Select the correct statement -

(1) Due to continental drift pouched mammals of Australia survived because of lack of competition from any other mammals.

(2) The study of geographical distribution of fauna and flora species in different parts of earth is called Biogerography.

(3) Whales, dolphins, like mammals live wholly in water.

(4) All of the above

85) Given below are some examples of mammals :-
Tiger cat, Banded anteater, Wombat, Koala, Lemur, Bobcat.
How many among these are not related to marsupial radiation ?

(1) Two

(2) Three

(3) Four

(4) One

86) In 1953, S.L. Miller, an _____ scientist created primitive earth like conditions in a laboratory scale. He created electric discharge in a closed flask containing CH_4, H_2, NH_3 and water vapour at _____.

(1) American, 800 K

(2) Russian, 800°C

(3) American, 800°C

(4) French, 800°C

87) _____ had talked of inheritable 'factors' influencing phenotype, Darwin either ignored these observations or kept silence.

(1) Hugo deVries
(2) Mendel
(3) Lamarck
(4) Oparin

88) When more individuals aquire peripheral character value at both ends of distribution curve, that condition is called.

(1) Disruption
(2) Directional changes
(3) Stabilisation
(4) Equilibrium

89) Which of the following man was with a brain size of 1400 cc, used hide to protect their body and burried their dead?

(1) Homo habilis
(2) Homo erectus
(3) Cromagnon man
(4) Neanderthal man

90) Similarities in proteins & genes performing a given function among diverse organisms give clues to _____ .
Complete the above statement by using correct option :

(1) Different ancestry
(2) Analogous organs
(3) Convergent evolution
(4) Common ancestry

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	3	1	2	1	1	2	2	1	2	1	1	1	4	3	4	4	3	4	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	1	3	1	1	3	2	1	3	2	2	3	1	4	4	3	3	2	2	3
Q.	41	42	43	44	45															
A.	4	3	3	4	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	4	3	1	1	2	3	3	1	1	4	3	2	3	1	2	1	3	3	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	3	1	3	4	4	1	4	1	2	3	3	2	3	3	1	4	2	2	1
Q.	86	87	88	89	90															
A.	1	1	4	4	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	1	4	2	4	4	1	1	2	3	3	4	3	1	1	3	1	1	3	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	3	3	3	1	3	2	3	2	2	3	3	1	4	4	3	3	1	1	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	1	2	3	4	1	4	2	1	3	4	2	2	2	4	2	4	1	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	1	4	2	4	4	4	4	4	4	4	2	1	1	3	4	1	2	2	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	3	1	4	1	3	2	1	4	4										

SOLUTIONS

PHYSICS

1) The minimum energy required for an electron to escape the metal surface is indeed the definition of the work function of the metals, which is correct statement. When a negatively charged electron leaves the neutral metal surface, the surface is left with net positive charge. This is also correct statement.

2)

Relation of stopping potential
$|V_{OC}| > |V_{OB}| > |V_{OA}|$
Greater stopping potential ⇒ greater frequency
⇒ Smaller wave length
$\lambda_C < \lambda_B < \lambda_A$
That means
$\lambda_B > \lambda_C$ but $\lambda_B < \lambda_A$
Higher intensity = Higher saturation current.
 Intensity of A is not less than C.

3) $P = n\frac{hc}{\lambda}$

Power of light $= \frac{15 \times 10^{-3} \times 10}{100} = 15 \times 10^{-4}\,W$

$n = \frac{P\lambda}{hc}$

$= \frac{15 \times 10^{-4} \times 440 \times 10^{-9}}{2 \times 10^{-25}}$ per sec

$= 330 \times 10^{13} \times 60$ per min

$= 1.98 \times 10^{17}$ per min

4) $K_{max} = \left[\frac{12400}{5000} - 1.2\right] eV$

$K_{max} = 1.28$ eV

and $K_{min} = 0$

On reaching Anode

$K'_{max} = (1.28 + 4) = 5.28 eV$

$K'_{min} = (0 + 4) = 4 eV$

5) $\frac{i_1}{i_2} = \left(\frac{r_2}{r_1}\right)^2 \Rightarrow \frac{1.4\mu A}{i_2} = \left[\frac{6}{2}\right]^2$

$\Rightarrow i_2 = \frac{1.4\mu_A}{9}$

= 0.15 μA

6) COLM

$$mv = \frac{h}{\lambda}$$

$$v = \frac{h}{\lambda m}$$

$$\frac{1}{\lambda} = R(1)^2 \left[\frac{1}{2^2} - \frac{1}{4^2}\right]$$

$$\frac{1}{\lambda} = \frac{3}{16}R$$

$$v = \frac{3}{16}\frac{Rh}{m}$$

7) $$K_{max} = \left[\frac{12400}{5000} - 2.44\right] eV$$

K_{max} = 0.04 eV

$$\lambda = \frac{12.27}{\sqrt{0.04}} = 6.135 \times 10^{-9} m$$

8) **Explain question:**

Calculate the energy of a photon whose wavelength is equal to the de Broglie wavelength of an electron accelerated by a 125 V potential.

Concept:

de Broglie wavelength of an electron and energy-wavelength relationship for a photon.

Solution:

$$\lambda_e = \frac{12.27}{\sqrt{V}} \text{Å}$$

$$\lambda_e = \frac{12.27}{\sqrt{125}} = \frac{12.27}{11.18} \text{Å} \approx 1.1 \text{Å}$$

$$E_p = \frac{hc}{\lambda}$$

$$E_p = \frac{12400}{1.1 \text{Å}} eV$$

$$E_p = \frac{12427}{1.1} eV$$

$$E_p \approx 11272.7 eV$$

$$E_p = 11.27 keV$$

Comparing with given options, the value is nearest to 11.3 keV.

Final answer: Option (2)

9) is $0 \le x \le 1$ $K_1 = TE - U_1 = 2E_0 - E_0 = E_0$

is $x > 1$ $K_2 = TE - U_2 = 2E_0 - \frac{E_0}{2} = \frac{3}{2}E_0$

$$\lambda \propto \frac{1}{\sqrt{KE}} \Rightarrow \frac{\lambda_1}{\lambda_2} = \sqrt{\frac{K_2}{K_1}} = \sqrt{\frac{3}{2}\frac{E_0}{E_0}}$$

$= \sqrt{3} : \sqrt{2}$

10) $A \rightarrow {}^{A-4}_{Z-2}B + {}^{4}_{2}He$

${}^{A-4}_{Z-2}B \rightarrow {}^{A-4}_{Z}C + 2e^{-}$

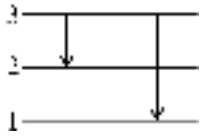
11) Q = [mass of reactant – mass of product] C^2

= [3 × 4.002604 – 12.00000] × 931.5 MeV

12) E_n = –13.6 + 12.5

= –1.1 eV

This energy level is nearly n = 3



so it may be lymen or Balmer series.

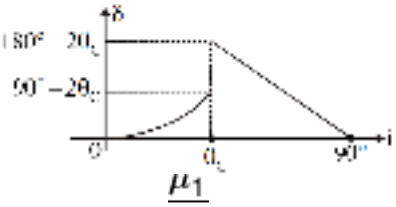
13) $v = \frac{Ze^2}{2\epsilon_0 hn} \Rightarrow v \propto \frac{1}{n}$

rectangular hyperbola.

14) $m = \frac{f}{f-u} = \frac{-f}{u-f} \Rightarrow m = -\frac{f}{x}$

15) Here, $\theta_C = 30°$

and $\sin\theta_C = \frac{\mu_r}{\mu_d}$



or $\sin 30° = \frac{\mu_1}{\mu_2}$

so, $\frac{\mu_1}{\mu_2} = \frac{1}{2}$

and $\delta_{max} = (180° - 2\theta_C) = 120°$ during TIR

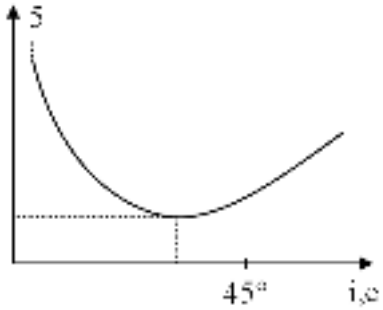
16) Detector will detect maximum light when the image of source is formed at the detector. Let this happens after a time t. At this time

u = – (90 – 10t); v = 10 t

from lens formula

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}; \frac{1}{10\,t} - \frac{1}{-(90-10\,t)} = \frac{1}{20}$

This gives t = 3 sec & t = 6 sec.



17)
$1 \sin i = \mu \sin (r_1)$
& $\mu \sin (r_2) = 1 \sin e$
& $r_2 + r_1 = A$
So, by decreasing i, e increases but $|\Delta i| \neq |\Delta e|$,
if $i > e$, then $|\Delta i| > |\Delta e|$
hence, new value of e will be
$37° < e < 40°$

18)

Visual angle of object and image is same
$\Rightarrow m = \frac{D}{|u|} = \frac{D}{|D|} = 1$

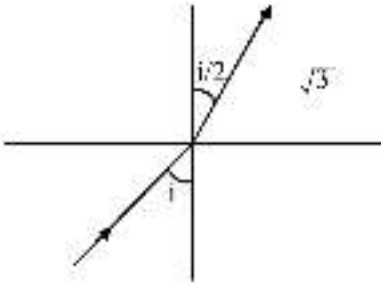
19)

In compound microscope objective forms image beynod $2f_0$.

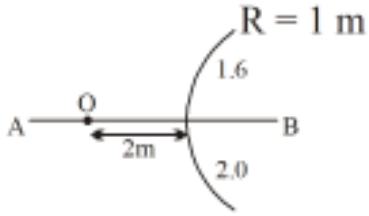
20) There are only 2 reflection if $\alpha > 60°$

21) $\frac{1}{-0.5} + \frac{1}{0.2} = \frac{2}{r} \Rightarrow \frac{2}{r} = 3$;
r = 0.667 m

22) $1 \sin i = \sqrt{3} \sin \frac{i}{2}$
$2 \cos \frac{i}{2} = \sqrt{3}$
$\cos \frac{i}{2} = \frac{\sqrt{3}}{2}$
$\frac{i}{2} = 30°$
$i = 60°$



23) $\mu \propto \frac{1}{\sin \theta}$



24)

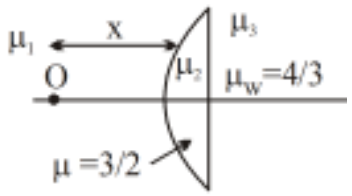
$$I_1 \Rightarrow \frac{1.6}{v_1} - \frac{1}{-2} = \frac{1.6-1}{+1} \Rightarrow v_1 = 16$$

$$I_2 \Rightarrow \frac{2.0}{v_2} - \frac{1}{-2} = \frac{2-1}{1} \Rightarrow v_2 = 4$$

$|v_1 - v_2| = 12$ m

25)

Converging lens form virtual enlarged image when object is placed between optical centre and focus.



26)

$$\frac{\mu_3}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R_1} + \frac{\mu_3 - \mu_2}{R_2}$$

$$\frac{\mu_3}{\infty} - \frac{1}{x} = \frac{1.5-1}{10} + \frac{4/3 - 3/2}{\infty}$$

$$\frac{-1}{x} = \frac{0.5}{10}$$

$x = -20$ cm

27)

$$m_1 - m_2 = \frac{d}{f}$$

$$2 - 0.5 = \frac{30}{f}$$

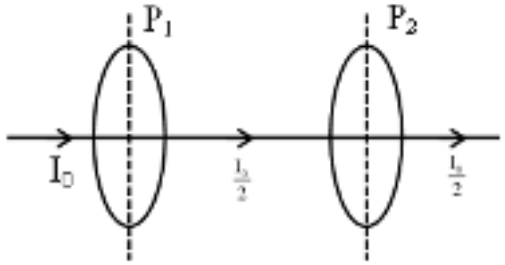
$f = 20$ cm

28) $f_0 + f_e = 42$

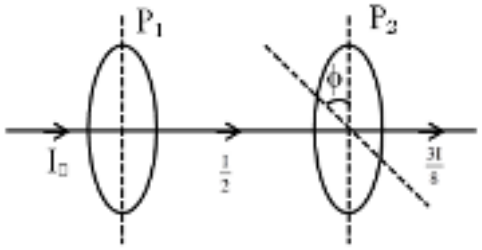
$$\frac{f_0}{f_e} = 6$$

29) Same intensity since $I_{min} = 0$

S_1, S_2 must not have equal phase but constant phase difference.

30) 

$I = \frac{I_0}{2}\cos^2\phi$



$\frac{I}{2}\cos^2\phi = \frac{3I}{8}$

$\cos\phi = \frac{\sqrt{3}}{2} \Rightarrow \phi = 30°$

31) $I_1 = I_2 = I'$

Maximum Intensity = 4I'

$I = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2}\cos\phi$

$I' = I' + I' + 2\sqrt{I'}\sqrt{I'}\cos\phi$

$I' = 2I' + 2I'\cos\phi$

$1 = 2(1 + \cos\phi)$

$\frac{1}{2} = 1 + \cos\phi$

$\cos\phi = -\frac{1}{2}$

$\phi = 120°$ or $\frac{2\pi}{3}$

$\Delta x = \frac{\lambda}{3}$

$\frac{\lambda}{3} = (\mu - 1)t$

$\frac{4000 \times 10^{-10}}{3} = (1.5 - 1)t$

$t = 2.67 \times 10^{-7}$ m

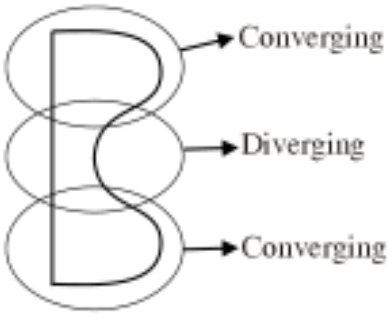
32) At path difference $\frac{\lambda}{6}$, phase difference is $\frac{\pi}{3}$

$I = I^0 + I^0 + 2I^0 \cos\frac{\pi}{3} = 3\ I^0$

$I_{max} = 4I_0$

So the required ratio is $\frac{3\ I_0}{4\ I_0} = 0.75$.

33)



(A)

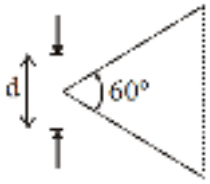


34) $\frac{2\lambda D}{a} = \frac{2 \times 600 \times 10^{-9} \times 2}{4 \times 10^{-4}} = = 0.006$ m

35) In diffraction

d sin 30° = λ

$\lambda = \frac{d}{2}$



Young's fringe width

[d' – separation between two slits]

$\beta = \frac{\lambda \times D}{d'}$

$10^{-2} = \frac{d}{2} \times \frac{50 \times 10^{-2}}{d'}$

$10^{-2} = \frac{10^{-6} \times 50 \times 10^{-2}}{2 \times d'}$

$\boxed{d' = 25\mu m}$

36)

In N-type S.C. $\sigma \approx n_e e \mu_e$

$6.24 \approx n_e \times 1.6 \times 10^{-19} \times 3900$

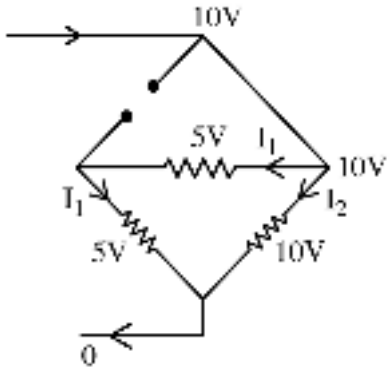
$n_e \approx 10^{16}/cm^3$

37)

If $V_P > V_N$ then p-n junction will be in forward bias.

38)

In each half cycle
One diode is F.B. and other is R.B.



39) $\frac{V_1}{V_2} = \frac{N_1}{N_2}$

$\frac{100}{V_2} = \frac{2}{1}$ $\Rightarrow V_2 = 50$ volt.

$(V_2)_{peak} = 50\sqrt{2}$ volt

$V_{DC} = \frac{V_{peak}}{\pi} = \frac{50\sqrt{2}}{\pi}$

40) (a) represents AND gate so → p

$\overline{\overline{A} + \overline{B}} = \overline{\overline{A}}.\overline{\overline{B}} = AB$

(b) represent OR gate so → r.

$\overline{\overline{A} \cdot \overline{B}} = \overline{\overline{A}} + \overline{\overline{B}} = A + B$

(c) represents NAND so → q

(d) represents NOR so → s

41) Slope = –1, From mirror equation $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\Rightarrow \frac{1}{v} = -\frac{1}{u} + \frac{1}{f}$ $\Rightarrow$ intercept $C = \frac{1}{f}$

$\Rightarrow f = \frac{1}{C} = \frac{1}{-0.05}$

$\Rightarrow f = -20cm \ : \ P = \frac{-100}{f(cm)}$

$= \frac{-100}{-20} = +5D$

42) $\delta_{min} = 2i - A \Rightarrow A = 2i - \delta_{min}$

$A = 2 \times 45° - 30° = 60°.$

$\mu = \frac{\sin\frac{(A+\delta min)}{2}}{\sin A/2} = \frac{\sin\left(\frac{60+30}{2}\right)}{\sin\left(\frac{60}{2}\right)} = \frac{1 \times 2}{\sqrt{2}} = \sqrt{2}$

$\mu = \sqrt{2} \Rightarrow \sin\theta_C = \frac{1}{\mu gm} = \frac{1}{\sqrt{2}}$

$\Rightarrow \theta_C = 45°$

43) Real thickness = c – a
Apparent thickness = c – b

$$\mu = \frac{c-a}{c-b}$$

$$= \frac{5.350-2.150}{5.350-3.350} = \frac{3.200cm}{2.000cm}$$

$$\Rightarrow \mu = \frac{3.200}{2.000} = 1.600$$

44) Dynamic resistance at 0.7 V

$$r_d = \frac{0.8-0.6}{(60-50)\times 10^{-3}} = 20\Omega$$

45) (a) First graph represent variation of current with applied bias for a normal diode, significant rise is observed in forward bias.
(b) Graph is for solar cell, which is obtained in III quadrant.
(c) This represents that current through capacitor decreases as potential difference across it increases.
(d) Fourth graph represents, Zener diode characteristics with sharp breakdown voltage in reverse bias.

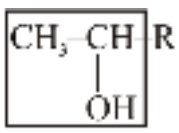
CHEMISTRY

46)

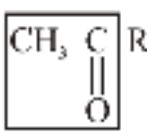
In PPS $RS^{\ominus}$ is better nucleophile than $RO^{\ominus}$

47)

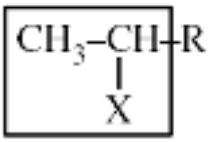
Conditions to give iodoform test



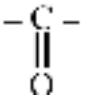
$CH_3–CH_2–OH$

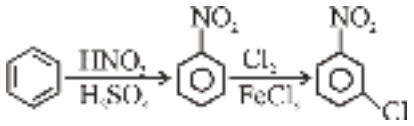


$CH_3–CHO$



48)

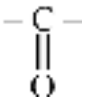
In haloform $–CH_3$ group of  converts into carboxylate ion

49) 

50)

Monosaccharides are reducing sugar.

51)

Rate of NAR ∝ +ve density on 'C' of  group.

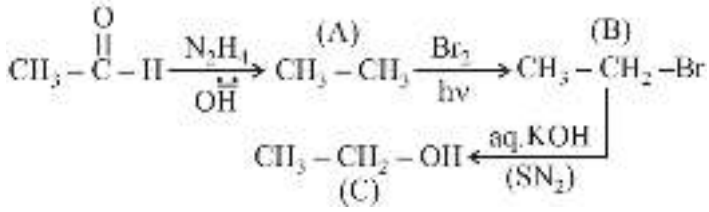
52)

Haloform reaction ⇒ A + B
Oxidation into acid → C

53) Fact

54) NCERT-XII, Pg # 372

55) NCERT-XII, Pg # 372

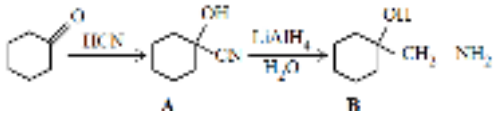
56) 

57)

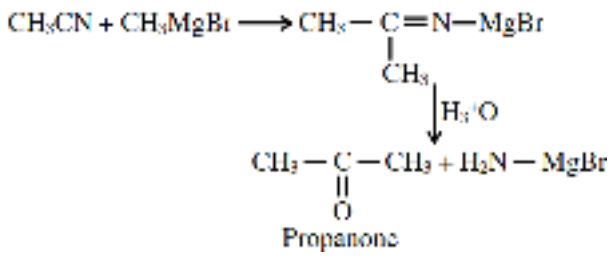
Ketone doesn't reduce tollen's reagent.

58)

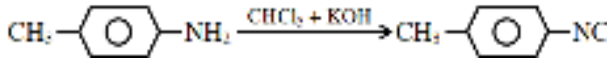
NCERT-XII, Pg.#365 (index Q.12.5)

59) 

60)



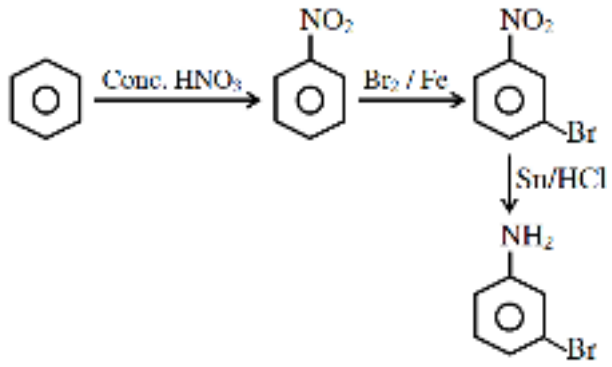
61)



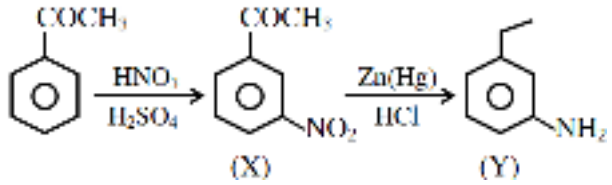
62)

R – X + KCN → R – CN
R – X + AgCN → R – NC

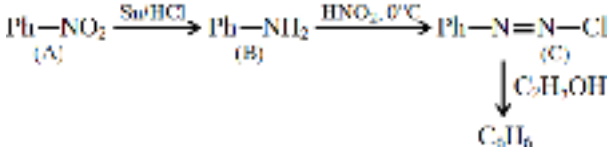
63)



64)



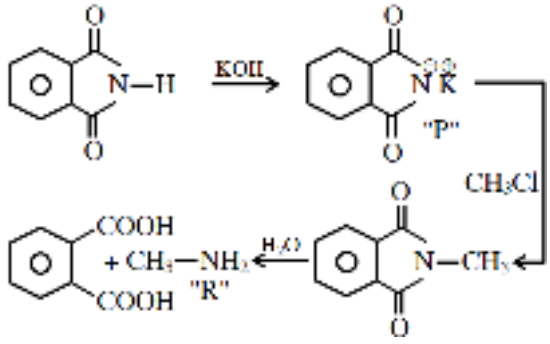
65)



66)

Only 1° amines give carbyl amine test.

67) In strongly acidic medium, aniline is protonated to form the anilinium ion which is meta directing.

68) 

69) CH_3–CH_2–NH_2 + HNO_2 → CH_2–CH_2–OH + N_2↑ + H_2O

70)

Sucrose is non reducing sugar.

71)

Monomer of cellulose is β-D-glucose.

72)

Ascorbic acid is vitamin 'C'.

73)

D-Mannose, D-Fructose, D-Glucose give same osazone.

74) NCERT (XIIth) Pg. # 389 - (Para13.3)

75)

w = 0.25 gm
V_1 = 40 ml
T_1 = 300 k
725mm aq. Tension = 25 mm
Act. pr. = 725 –25 = 700 mm

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\Rightarrow V_2 = \frac{700 \times 40}{300} \times \frac{273}{760} = 33.526ml$$

Now
 22400ml —→ 28g

$$1 \text{ ml} \quad —→ \quad \frac{28}{22400}$$

$$33.526 \text{ ml} \quad —→ \quad \frac{28}{22400} \times 33.526$$

= 0.0419 weight of N_2

$$\% N = \frac{0.0419}{0.25} \times 100 = 16.76\%$$

76)

Wurtz reaction followed by aromatization.

77) 11th NCERT (12.8.5) chromatography

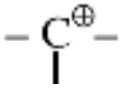
78)

$Fe_4[Fe(CN)_6]_3$
prussian blue

79)

Aryl halides undergo NSR at high temperature.

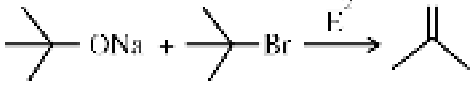
80)

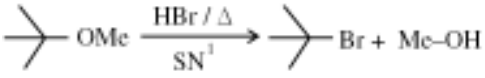
$S_N1 \propto$ stability of 

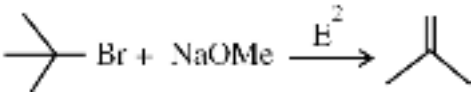
81)

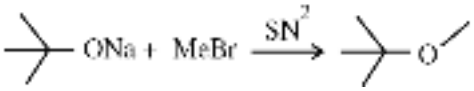
In E^2 rearrangement doesn't take place.

82)

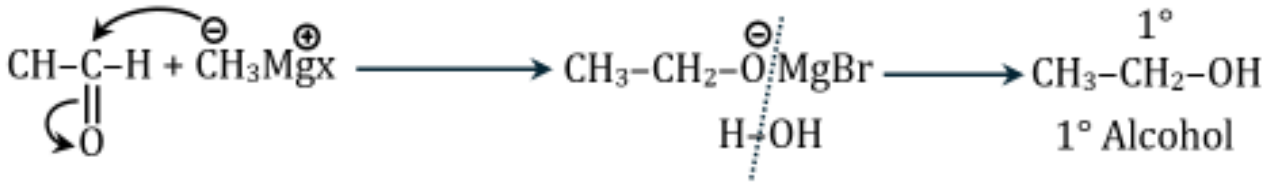
(A) 

(B) 

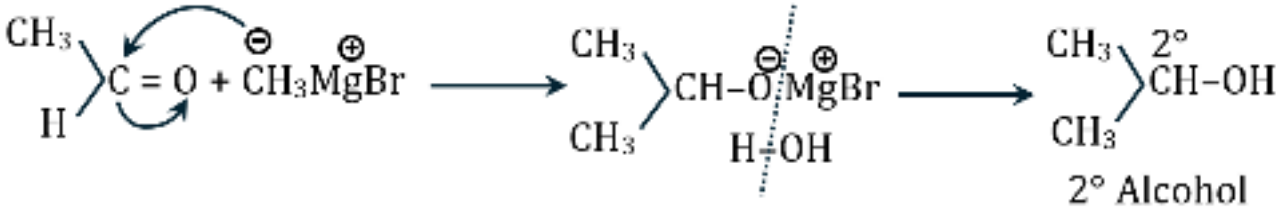
(C) 

(D) 

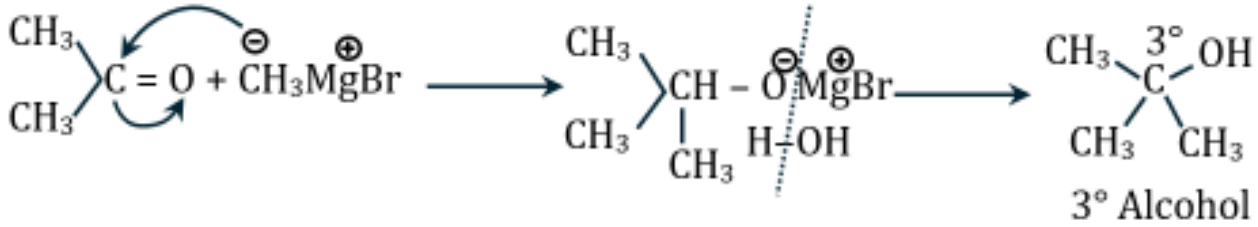
83) **Explaination:** Find out the reaction in which 2° Alcohol is obtained as a product.
Concept: NAR of carbonyl compounds.
Solution:
(1)



(2)

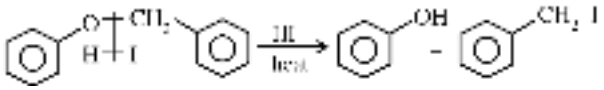


(3)

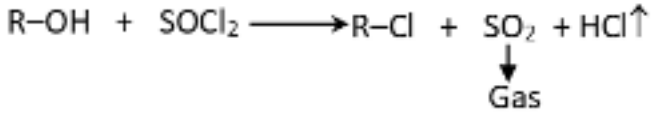


So, option (2) reaction give 2° Alcohol.
Final Answer: Option (2)

84)

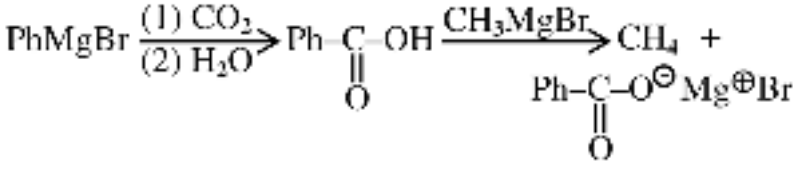


85) **Concept :**



Due to removal of SO_2(gas) and fumes of HCl, their reaction is thermodynamically more favourable ($\Delta S > 0$)
Option Ans. (1)
$ROH+SOCl_2 \rightarrow R\text{–}Cl+SO_2+HCl$

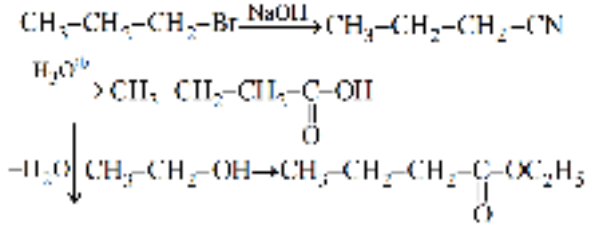
86)



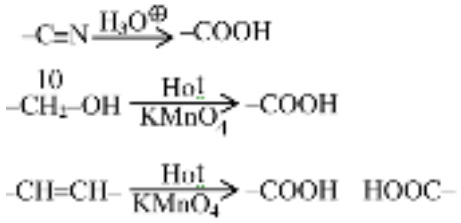
87)

Esterification $\propto \frac{1}{\text{Steric hinderance}}$

88)



89)



90)

NCERT XII Pg. # 397, 398
Part II

BIOLOGY

91) NCERT XII, Pg. # 207

92) NCERT XII, Pg. # 209

93) NCERT XII, Pg. # 210

94) NCERT XII, Pg. # 212

95) All statements are correct.

96) NCERT XII, Pg. # 206, 207

97) NCERT XII, Pg. # 213

98) NCERT XII, Pg. # 206
Decomposition, Nutrient cycling, productivity

99) NCERT XII, Pg. # 208

100) NCERT XII, Pg. # 221

101) NCERT XII, Pg. # 207

102) NCERT XII, Pg. # 244–246

103) NCERT XII, Pg. # 197 to 202

104) NCERT XII, Pg. # 206,207

105) NCERT XII, Pg. # 198,199

106) NCERT XII, Pg. # 202, 198

107) NCERT XII, Pg. # 198

108) NCERT XII, Pg. # 198

109) NCERT XII, Pg. # 201

110)

NCERT XII, Pg. # 221

111) NCERT XII, Pg. # 199

112) NCERT XII, Pg. # 199

113) NCERT XII, Pg. # 194

114)

NCERT XII, Pg. # 198,199

115) NCERT XII, Pg. # 197-201

116) NCERT Pg. # 201, 202, 207

117) **Explain Question:** The question is asking you to look at the table of effects (+, –, 0) and decide which statements (i–iv) correctly identify the types of interactions (A, B, C, D) shown in the table.
Concept: Population Interaction

Solution:
The correct answer is Option 2.
We analyze interactions A, B, C, and D based on the table:
A = (Harm, Benefit) Organism 1 harmed (–), Organism 2 benefited (+).
This pattern fits: Predation or Parasitism
So Statement (i) is incorrect.
B = (Benefit, No Effect)
Organism 1 benefits (+),
Organism 2 not affected (0)
→ This is Commensalism,
So Statement (ii) is correct.
C = (No Effect, Harm) Organism 1 not affected (0),
Organism 2 harmed (–)
→ This is Amensalism,
So Statement (iii) is correct.
D = (Harm, Harm) Both harmed (–, –)
→ This is Competition,
So Statement (iv) is incorrect.
Therefore the correct statements are: (ii) and (iii)
Which matches: Option 2
Final answer: The correct answer is option 2. (ii) and (iii)

118) NCERT XII, Pg. # 220

119) NCERT XII, Pg. # 217

120) NCERT XII, Pg. # 221

121) NCERT XII, Pg. # 219

122) NCERT XII, Pg. # 225

123) NCERT XII, Pg. # 219, 220

124) NCERT XII, Pg. # 219

125) NCERT-XII, Pg # 222-223

126) NCERT-XII, Pg # 218

127) NCERT XII, Pg. # 223

128) NCERT XII, Pg. # 224

129) NCERT XII, Pg. # 223

130) NCERT XII, Pg. # 223

131) NCERT XII, Pg. # 195

132) NCERT XII, Pg. # 193

133) NCERT-XII, Pg # 197

134) NCERT-XII, Pg # 192

135) NCERT XII, Pg. # 192,194,195

136) NCERT XII Pg. No. # 136

137) NCERT Pg.#152

138) NCERT, Pg # 147(E), 158(H)

139)

NCERT XII Page No. # 138

140) NCERT Pg. # 131, 133

141)

NCERT XII Page No. # 141, 142

142)

NCERT XII Page No. # 134

143) NCERT Pg#159

144) NCERT Pg.# 138

145) NCERT Pg. # 132-133

146) NCERT Pg. # 131

147) NCERT Pg. # 142,143

148) NCERT Pg. # 134, 136

149) NCERT-XII ; Page No. # 151

150) NCERT PAGE (ENGLISH): 143-146

151) NCERT Pg. # 137,138

152) NCERT-XII, Pg. # 130

153) NCERT Pg. # 160

154) NCERT Pg # (ENG) : 144

155) NCERT XII Pg. No. # 143

156) NCERT Pg # (ENG) : 142

157) NCERT XII, Pg. # 140

158) NCERT Pg. # 134

159) NCERT Page No.: 131(E), 143(H)

160) NCERT Pg # (ENG) : 137

161) NCERT-XII, Pg. No. # 137

162) NCERT-XII, Pg. # 124

163) NCERT Pg. No. # 118, 119

164) NCERT Pg. No. # 117, 118

165) NCERT XII Pg. No. # 111

166) NCERT Pg.# 111, 112.

167) NCERT XII Pg. No. # 111

168) NCERT PAGE (ENGLISH): 115,116

169) NCERT Pg. # 118

170) NCERT XII Pg. No. # 114

171) NCERT Pg. # 122-124/133-134_(H)

172) NCERT XII Pg. No. # 112, 113

173) NCERT Pg. # 110

174) NCERT Pg. # 124

175) NCERT Pg. # 117/129_(H)

176) NCERT PAGE (ENGLISH): 111

177) NCERT XII Pg. No. # 119

178) NCERT Page No.: 120(E), 132(H)

179) NCERT Page No.: 125

180) NCERT Pg. No. # 115